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Enteral lipid supplements for the prevention and treatment of parenteral nutrition-associated liver disease in infants (Review)

Premkumar MH, Huff KA, Cooper C, Cracknell J, Pammi M, supported by the Cochrane Neonatal Review Group

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[Intervention Review]

Enteral lipid supplements for the prevention and treatment of parenteral nutrition-associated liver disease in infants

Muralidhar H Premkumar[1], Katie A. Huff[2,3,4], Chris Cooper[5], Jane Cracknell[6], Mohan Pammi[1], supported by the Cochrane Neonatal Review Group

[1]Division of Neonatology, Department of Pediatrics, Baylor College of Medicine, Houston, Texas, USA. [2]Department of Pediatrics, Indiana University School of Medicine, Indianapolis, Indiana, USA. [3]Department of Pediatrics, University of Cincinnati College of Medicine, Cincinnati, Ohio, USA. [4]Division of Neonatology, Cincinnati Children's Hospital Medical Center, Cincinnati, Ohio, USA. [5]Population Health Sciences, Bristol Medical School, University of Bristol, Bristol, UK. [6]Cochrane Neonatal, Vermont Oxford Network, Oxford, UK

Contact: Muralidhar H Premkumar, premkuma@bcm.edu.

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ABSTRACT

Rationale

Enteral lipid formulations may offer benefits according to preclinical studies, but their effectiveness and safety in preventing and treating parenteral nutrition-associated liver disease (PNALD) in infants remains unknown.

Objectives

To evaluate the benefits and harms of enteral lipid supplementation for the prevention and treatment of PNALD in infants.

Search methods

We searched CENTRAL, MEDLINE, Embase and trial registers, together with reference checking and contact with study authors. The latest search was in December 2024.

Eligibility criteria

We included parallel, randomized control trials (RCTs) in infants comparing enteral lipid supplements with either a placebo or no intervention in infants who were either at risk or who had PNALD.

We excluded studies involving infants receiving enteral feeds without the need for parenteral nutrition, as they were neither at risk for PNALD, nor diagnosed with PNALD, and infants with cholestasis secondary to primary liver disease, inborn errors of metabolism, or congenital infection.

Outcomes

Our primary outcomes were the prevention of PNALD during hospital stay and the first year of life; and resolution of PNALD during hospital stay and the first year of life. Other outcomes included feeding intolerance during the hospital stay, time to full feeds measured in days during the hospital stay, length of hospital stay in days, need for liver transplantation in the first year of life and all-cause mortality up to hospital discharge and in the first year of life.

Risk of bias

We used the Cochrane risk of bias 1 (RoB 1) tool to assess bias in the RCTs.

Synthesis methods

We synthesized results for each outcome using meta-analyses where possible, using random-effect models to calculate risk ratios (RRs) and risk difference (RD) with 95% confidence intervals (CIs) for dichotomous outcomes. For continuous outcomes, we calculated mean differences (MDs). If combining the data was not feasible, we summarized data narratively using Synthesis without Metaanalysis (SWiM).

We summarized the certainty of evidence according to GRADE methods.

Included studies

We included 11 studies (2192 infants). A variety of enteral lipids were used as interventions, including algal oil, fungal oil, and fish oil, either alone or in combination. The intervention groups received:

- Fish oil alone or in combination with other oils (three studies; n = 1328);

- Algal oil alone or in combination with other oils (eight studies; n = 864).

The comparison groups received either no intervention or other oils, such as sunflower oil, safflower oil, olive oil, or medium-chain triglyceride (MCT) oil.

Most studies included preterm infants who received enteral lipid supplements while primarily on parenteral nutrition and minimal enteral nutrition, thus being considered at risk for PNALD. No studies examined enteral lipid supplementation in infants with established PNALD.

Synthesis of results

Any enteral lipids compared to placebo or no treatment in infants at risk or with PNALD

Enteral lipids may have little to no effect on the prevention of PNALD as measured by incidence of cholestasis during the hospital stay and in the first year of life; however, the evidence is very uncertain (RR 0.65, 95% CI 0.13 to 3.14; 3 studies, 265 participants; very low-certainty evidence). Enteral lipids may have little to no effect on feeding intolerance, but the evidence is very uncertain (RR 0.14, 95% CI 0.02 to 1.12; 4 studies, 236 participants; very low-certainty evidence). The evidence suggests that enteral lipids result in little to no difference in time taken to achieve full enteral feeds, but the evidence is very uncertain (MD −1.10, 95% CI −2.98 to 0.78; 1 study, 1273 participants; very low-certainty evidence). Enteral lipid supplementation may decrease the length of hospital stay, but the evidence is very uncertain (MD −3.44, 95% CI −6.20 to −0.68; I^2 not applicable; 1 study, 67 participants; very low-certainty evidence). Enteral lipids may have little to no effect on mortality up to discharge and the first year of life, but the evidence is very uncertain (RR 1.19, 95% CI 0.90 to 1.57; I^2 = 0%; 11 studies, 2200 participants; very low-certainty evidence).

No studies reported on the resolution of PNALD during the hospital stay and in the first year of life, or the need for liver transplantation.

Limitations of evidence: a wide variety of enteral lipids in both intervention and control groups made it challenging to make comparisons between the studies. Overall, the risk of selection bias was unclear to high and the certainty of evidence very low. The certainty of evidence was downgraded due to several methodological limitations, including risk of bias (allocation, blinding, and performance/detection issues), indirectness from the exclusion of high-risk infants, inconsistency in outcome definitions, and imprecision from small samples and wide confidence intervals. Incomplete reporting across studies also raised concerns about publication bias.

Authors' conclusions

The evidence is very uncertain about the effect of enteral lipids on the prevention of PNALD during the hospital stay and in the first year of life. There may be little to no difference in feeding intolerance or time to achieve full enteral feeds. They may decrease the length of stay and likely do not reduce or increase mortality up to discharge or during the first year of life in infants at risk for PNALD.

Due to small samples, inconsistency, and risk of bias, the available data are of very low certainty, and so we are not able to draw conclusions about the effects of enteral lipids in infants who are at risk of PNALD. No studies evaluated resolution of PNALD during the hospital stay and in the first year of life, or the need for liver transplantation.

Well-designed studies are needed to evaluate the short-term and long-term effects of different enteral lipids in infants who are at risk for PNALD or have PNALD. Future studies should focus on addressing the heterogeneity of enteral lipid preparations and the inconsistency in assessment of outcomes.

Funding

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Registration

Protocol (2021) DOI: 10.1002/14651858.CD014353.

PLAIN LANGUAGE SUMMARY

What are the benefits and risks of oils given by mouth (enteral lipids) in infants who either have or are at risk of liver disease due to intestinal failure?

Key messages

- As there is no good-quality evidence, we do not know if enteral lipids cause benefits or harms in infants at risk of liver disease or who have liver disease due to prolonged need for artificial nutrition secondary to intestinal failure.
- There were no studies that specifically studied the effect of enteral lipids in infants at risk of liver disease or who have liver disease due to prolonged need for artificial nutrition secondary to intestinal failure.
- To answer this question, we need large studies specifically designed to study the effect of enteral lipids in infants with intestinal failure.

What is parenteral nutrition-associated liver disease?

Infants (children younger than one year) with intestinal failure, who cannot be fed by mouth or a feeding tube, rely on parenteral nutrition (artificial nutrition given through a vein). A common complication of this treatment is parenteral nutrition-associated liver disease (PNALD), which can lead to severe illness or even death if not properly managed. The exact causes of this liver disease are unclear, but contributing factors include artificial nutrition, lack of oral feeding, infections, and surgeries. Currently, there are no fully effective ways to prevent liver disease in infants with intestinal failure. Some types of artificial nutrition, such as those containing fish oil, may help, but they do not completely prevent or treat liver disease in infants with intestinal failure. Therefore, better prevention and treatment strategies are needed.

What are enteral lipids?

Lipids (fats and oils) are essential nutrients that provide energy, transport vitamins, and serve as building blocks for cells. They are also critical for brain and eye development. When lipids are given by mouth or through a feeding tube, they are called enteral lipids. These lipids can come from various sources, including fish, plants, algae, fungi, or synthetic production, and their composition varies depending on the source. Laboratory studies in animals suggest that enteral lipids may help prevent or reduce liver disease in the setting of intestinal failure.

What did we want to find out?

We wanted to know if enteral lipids provide benefits or cause harm in infants at risk of liver disease or who have liver disease due to prolonged need for artificial nutrition secondary to intestinal failure. Specifically, we asked whether enteral lipids could prevent or treat liver disease in infants with intestinal failure.

What did we do?

We searched the medical literature for studies evaluating the benefits and risks of enteral lipids in infants at risk for or already diagnosed with PNALD. We included only studies where enteral lipids were given by mouth or through a feeding tube, and where infants were assigned by chance to different treatment groups. We then compared, analyzed, and summarized the results of these studies, assessing our confidence in the evidence, based on factors such as study methods and the number of infants involved.

What did we find?

We found 11 studies involving 2192 infants at risk of PNALD who received enteral lipids for up to approximately eight weeks. Of these studies, four compared algal oil alone or in combination with other oils to a control group, while three studies compared fish oil alone or in combination with other oils to a control group. The smallest study included 18 infants, while the largest included 1273. Studies were conducted across Asia, Australia, Europe, and the Americas. Several studies were supported by pharmaceutical companies.

Only a few studies assessed the prevention of PNALD. We did not find any studies evaluating enteral lipids (enterally) for the treatment of established liver disease. Eleven studies investigated enteral lipids in infants at risk of liver disease. We found that, compared to no treatment or placebo (no active substance), enteral lipids may have little to no effect on the prevention of: PNALD; feeding intolerance; time to achieve full enteral feeds by mouth or a feeding tube; or mortality. The length of hospital stay may be decreased with enteral lipids, but there is limited information available from only a single study.

Overall, we are uncertain about these results.

No data evaluated the effect of enteral lipids on the need for liver transplantation.

What are the limitations of the evidence?

We are not confident that enteral lipids have any effect on PNALD, feeding intolerance, the time to achieve full enteral feeds, length of hospital stay, or mortality because:

1. The studies varied in the type, dose, duration, and administration of enteral lipid;

2. Some study participants or researchers may have been aware of which treatment they were receiving;
3. Not all studies provided data on all outcomes of interest;
4. There were too few studies to draw definitive conclusions, and some were very small.

How up-to-date is this evidence?

The evidence is up-to-date to 16 December 2024.

SUMMARY OF FINDINGS

Summary of findings 1. Summary of findings table - enteral lipid supplements vs. placebo or no treatment in infants at risk for or with parenteral nutrition-associated liver disease

Patient or population: infants at risk for or with parenteral nutrition-associated liver disease
Setting: Neonatal intensive care units, pediatric intensive care units, and pediatric units in Asia, South America, and North America
Intervention: enteral lipid supplements
Comparison: placebo or no treatment

Outcomes	Anticipated absolute effects* (95% CI)		Relative effect (95% CI)	№ of participants (studies)	Certainty of the evidence (GRADE)	Comments
	Risk with placebo or no treatment	Risk with enteral lipid supplements				
Prevention of parenteral nutrition-associated liver disease during the hospital stay and in the first year of life (incidence of cholestasis) assessed with: serum bilirubin levels follow-up: range 2 weeks to 17 weeks	82 per 1000	**53 per 1000** (257 to 11)	**RR 0.65** (0.13 to 3.14)	265 (3 RCTs)	⊕⊝⊝⊝ Very low[a]	Enteral lipids may have little to no effect on the risk of developing PNALD as measured by the incidence of cholestasis during the hospital stay and in the first year of life, however the evidence is very uncertain.
Resolution of parenteral nutrition-associated liver disease - not measured	-	-	-	-	-	No studies were found that measured the resolution of parenteral nutrition-associated liver disease.
Feeding intolerance assessed with: clinical observation follow-up: range 14 days to 56 days	65 per 1000	**9 per 1000** (73 to 1)	**RR 0.14** (0.02 to 1.12)	236 (4 RCTs)	⊕⊝⊝⊝ Very low[b]	Enteral lipids may have little to no effect on feeding intolerance but the evidence is very uncertain.
Time taken to achieve full enteral feeds (in days) during the hospital stay (time to full feeds) assessed with: clinical assessment	The mean time taken to achieve full enteral feeds (in days) during the hospital	MD **1.1 days fewer** (2.98 fewer to 0.78 more)	-	1273 (1 RCT)	⊕⊝⊝⊝ Very low[c]	Enteral lipids may have little or no effect on the time taken to achieve full enteral feeds but the evidence is very uncertain.

	stay was **18.9** days					
Length of stay (hospital stay)	The mean length of stay was **25.07** days	MD **3.44 days fewer** (6.2 fewer to 0.68 fewer)	-	67 (1 RCT)	⊕⊝⊝⊝ Very low[d]	Enteral lipids may decrease the length of hospital stay, but the evidence is very uncertain. The true effect may be substantially different from the estimate of the effect.
Need for liver transplantation in the first year of life - not measured	-	-	-	-	-	No studies were found that measured the need for liver transplantation.
All-cause mortality up to hospital discharge and the first year of life (death)	62 per 1000	**74 per 1000** (98 to 56)	**RR 1.19** (0.90 to 1.57)	2200 (11 RCTs)	⊕⊝⊝⊝ Very low[e]	Enteral lipids may have little to no effect on mortality up to discharge or during the first year of life, but the evidence is very uncertain.

*__The risk in the intervention group__ (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

CI: confidence interval; **MD:** mean difference; **RR:** risk ratio

GRADE Working Group grades of evidence
High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.
Moderate certainty: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.
Low certainty: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.
Very low certainty: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

See interactive version of this table: https://gdt.gradepro.org/presentations/#/isof/isof_question_revman_web_453219006472731855.

[a] Downgraded three levels for a. risk of bias: concerns of allocation and blinding in Hellstrom 2021 study; b. indirectness: the eligible studies did not include high-risk groups such as infants with intestinal failure; c. imprecision: small number of infants in two out of three studies (Frost 2021 and Robinson 2016) and wide confidence intervals of the overall effect

[b] Downgraded three levels for a. inconsistency: since vomiting is just one of the several ways of assessing feed intolerance across studies; b. indirectness: due to lack of infants with prolonged parenteral nutrition in the study; c. imprecision: serious concerns for a single study with small numbers and wide confidence intervals of the overall effect

[c] Downgraded three levels for: a. indirectness: the study population did not have prolonged need for parenteral nutrition; b. imprecision: due to a single study with wide confidence intervals; c. publication bias: the lack of reporting of this outcome in other studies raises concerns for publication bias

[d] Downgraded three levels due to: a. risk of bias: due to performance and detection bias; b. indirectness: the eligible studies did not include high-risk groups such as infants with intestinal failure; c. imprecision: due to a single study with small numbers and wide confidence intervals; d. publication bias: the lack of reporting of this outcome in other studies raises concerns for publication bias

[e] Downgraded three levels because of: a. risk of bias: concerns for randomization and allocation concealment; b. imprecision: the pooled estimate effect of the seven studies had small numbers and wide confidence intervals c. inconsistency: though statistical heterogeneity was low, this instance of downgrading is due to the noted clinical and methodological heterogeneity; d. indirectness: due to lack of infants with prolonged parenteral nutrition in the included studies

BACKGROUND

Description of the condition

Parenteral nutrition-associated liver disease (PNALD), also known as parenteral nutrition-associated cholestasis or intestinal failure-associated liver disease, is a hepatobiliary dysfunction caused by intestinal failure [1]. It is diagnosed by conjugated bilirubin ≥ 2 mg/dL in patients receiving prolonged parenteral nutrition, excluding other cholestasis causes. Up to 50% of infants on long-term parenteral nutrition develop PNALD [2, 3], with incidence rising to 80% in low-birth-weight infants with intestinal failure post-intestinal resection [4]. Histologic and biochemical signs of PNALD may appear as early as 14 days after onset of intestinal failure [5, 6].

PNALD has a multifactorial etiology, including prolonged parenteral nutrition, lack of enteral feeding, prematurity, infections, abdominal surgery, and small intestinal bacterial overgrowth [2, 3, 7, 8, 9, 10]. Intravenous lipid emulsions, essential in providing lipid nutrition to these infants, are implicated in PNALD pathogenesis. Soybean oil-based lipid emulsions, standard since 1962, have been associated with PNALD when used long term [3, 8, 11]. Their hepatotoxicity is linked to high levels of stigmasterol and pro-inflammatory omega-6 fatty acids, alongside low concentrations of vitamin E and DHA [11, 12]. Newer lipid emulsions, including 100% fish oil-based and mixed-oil formulations, offer alternatives that may prevent or treat PNALD [8, 9, 13, 14, 15]. These newer emulsions lack stigmasterol, are rich in vitamin E and polyunsaturated fatty acids, and possess anti-inflammatory omega-3 profiles. However, they are not always effective in the prevention and treatment of PNALD, and their use requires venous access, which carries risks of thromboembolism and bloodstream infections.

Description of the intervention and how it might work

Lipids are calorie-dense macronutrients, and are essential for growth, cell membrane stability, nerve conduction, and fat-soluble vitamin transport. Enteral lipids, delivered via the gastrointestinal tract, are absorbed primarily into the jejunum and duodenum after emulsification, entering lymphatic and portal circulation [16, 17, 18, 19]. Sources include plant oils (rich in n-6 and n-9 fatty acids), coconut oil (medium-chain triglycerides), and fish oil (long-chain polyunsaturated fatty acids) [20].

Enteral lipid supplements containing fish oil can improve cholestasis through both direct and indirect mechanisms [20, 21]. Rich in long-chain omega-3 polyunsaturated fatty acids, fish oil enhances intestinal adaptation in animal models and infants with intestinal failure. These supplements offer anti-inflammatory benefits and may support the resolution of PNALD, similar to fish oil-based intravenous lipid emulsions.

Therapeutically, enteral lipids aid nutrient absorption, promote gastrointestinal mucosal growth and motility, and support liver function. Their isosmotic nature reduces fluid loss and dehydration risk [22, 23]. It also reduces diarrhea, enhances fat absorption, and increases mucosal protein content, acting as a trophic factor [24, 25]. In ileocecal resection models, enteral fish oil improved fat absorption [26]. Fish oil increases peptide YY, slowing gastrointestinal transit and enhancing nutrient contact with mucosa [27, 28]. Compared to parenteral lipids, enteral lipids reduce liver fat accumulation, normalize serum transaminases, and improve intestinal adaptation [24, 29, 30]. Enteral fish oil also stimulates bile flow and enhances enterohepatic circulation, reducing reliance on soybean oil-based emulsions, which are linked to PNALD [21].

Although newer intravenous lipid emulsions help, they don't fully resolve PNALD, highlighting the need for complementary strategies like enteral lipids. Enteral lipids may improve digestion, absorption, and feed tolerance, accelerating the transition to full enteral nutrition. They also help reduce intestinal dysbiosis and inflammation. Thus, by improving feed tolerance and reducing parenteral nutrition dependence, enteral lipids could potentially decrease sepsis risk, central line use, the need for liver transplant, and ultimately improve survival in infants with PNALD.

Why it is important to do this review

Enteral lipids are a cost-effective, accessible option that may reduce PNALD-related complications [21, 31]. In the short term, they could enhance feeding tolerance, decrease reliance on parenteral nutrition, and complement therapies for PNALD prevention and treatment, potentially reducing long-term consequences. However, their role in either the prevention or treatment of PNALD remains underexplored.

At the 2022 FDA Workshop on Phytosterols in PNALD, experts stressed improving feeding tolerance and reducing parenteral nutrition dependence to lower PNALD incidence and severity [32]. In this context, investigating enteral lipids is timely. Yet concerns remain regarding their nutritional adequacy and their impact on short-term outcomes, including mortality. Despite their promise, no systematic review has yet assessed the benefits and harms of enteral lipid supplementation in infants at risk of PNALD or with PNALD. This review aims to address that gap.

OBJECTIVES

To evaluate the benefits and harms of enteral lipid supplementation for the prevention and treatment of PNALD in infants.

METHODS

We followed the Methodological Expectations for Cochrane Intervention Reviews (MECIR) when conducting the review [33], and PRISMA 2020 for the reporting [34, 35].

Since the publication of the protocol in 2021 [36], some methods have been updated as mentioned below.

1. We used Covidence for screening;
2. We clarified which participants were excluded;
3. We excluded the studies in which enteral lipids were given to infants who were on full enteral feeds. We adopted this approach since infants on full enteral feeds were, by definition, not on parenteral nutrition, hence not at risk for PNALD;
4. Due to the unexpected complexity of the types of enteral lipids studied in infants, we made the following modifications to lipid categorization:
    a. We broadened the categories of fish oil and plant-based oil to any fish oil and any plant-based oil to account for the variety within each original category;

b. We identified studies using algal and fungal oils that did not fit into the predefined categories of fish oil or plant-based oils. To account for this, we created a new category for other enteral lipids that did not belong to either group and included them in our comparisons;

5. To provide a more comprehensive analysis, we combined all eligible studies on fish oil and algal oil, comparing any enteral lipid formulation versus no treatment or placebo;
6. To address the heterogeneity between studies and concerns about the influence of small-study effects on the results of the meta-analysis, we chose random-effects analysis instead of fixed-effect analysis;
7. Two new authors joined the review team: Chris Cooper and Jane Cracknell.

Criteria for considering studies for this review

Types of studies

Randomized control trials (RCTs), quasi-RCTs and cluster-RCTs were eligible for inclusion in this review. Quasi-RCTs are parallel-arm RCTs in which allocation is decided by an approximation of randomization (e.g. allocation by patient ID number).

Types of participants

We included the following participants:

1. Infants at risk of parenteral nutrition-associated liver disease; and
2. Infants with parenteral nutrition-associated liver disease.

PNALD was defined as the elevation of serum levels of conjugated bilirubin over 2 mg/dL when exposed to parenteral nutrition for two weeks or more, in the absence of other causes of cholestasis [1, 2].

We defined an infant as a child under the age of one year.

We excluded infants with cholestasis due to primary liver disease, inborn errors of metabolism, or congenital infection, as their pathogenesis differs from PNALD and requires disease-specific treatments.

We excluded studies involving infants receiving enteral feeds without the need for parenteral nutrition, as they were neither at risk for PNALD, nor diagnosed with PNALD.

Had we identified studies including both eligible and ineligible populations, we would have extracted data only for the eligible populations based on data in the study, or based on information obtained from the study authors.

Types of interventions

We included studies of enteral administration of lipid supplements of any formulation compared to a placebo or no treatment, or against each other, at any dosage or duration. Lipid supplements include oils derived from various sources such as fish oil (tuna oil, salmon oil), plant-based oil (olive oil, sunflower oil), algal oil, fungal oil or mixed oil emulsions (algal and fungal oil, fish and medium-chain triglyceride oil).

We included the following comparisons.

1. Any fish oil versus no treatment or placebo;
2. Any plant-based oil versus no treatment or placebo;
3. Mixed oil emulsion versus no treatment or placebo;
4. Other enteral lipid (not defined above) versus no treatment or placebo;
5. One enteral lipid emulsion versus another enteral lipid emulsion;
6. Any enteral lipids versus no treatment or placebo.

Outcome measures

We included trials for which the study design, participants, interventions, and outcomes fulfilled the Criteria for considering studies for this review. We included studies which planned to measure outcomes of interest even if they reported no data for that outcome.

The critical and important outcomes in this review include core outcomes in PNALD [1].

Critical outcomes

1. Prevention of PNALD during the hospital stay and in the first year of life (PNALD is defined as the elevation of serum conjugated bilirubin of 2 mg/dL or greater for more than two weeks in the absence of other causes of cholestasis, or as defined by the investigators) [2].
2. Resolution of PNALD during the hospital stay and in the first year of life (defined as a sustained reduction of elevated conjugated bilirubin levels to below 2 mg/dL or less than 20% of the total bilirubin concentration for two weeks or more, or as defined by the investigators).

Important outcomes

1. Feeding intolerance is defined by the number of days when feeds were stopped or reduced and parenteral nutrition was either commenced or increased during the hospital stay secondary to the inability to digest enteral feeds presented as gastric residual volume of more than 50%, abdominal distension or emesis or both, or as defined by study authors [37].
2. Time taken to achieve full enteral feeds (in days) during the hospital stay is defined as the first day when 100% of the total caloric requirement or 140 mL/kg/day or greater of feeds was administered enterally and thereafter sustained for more than 48 hours or as defined by the authors [38].
3. Adequacy of growth based on:
    a. Rates of increase in body weight (grams/day or grams/kilogram/day), length (centimeters/week), and head circumference (centimeters/week) up to six months post-term;
    b. Long-term growth: rates of increase in body weight (grams/day), length (centimeters/week), and head circumference (centimeters/week) from six months post-term;
    c. Weight (grams or z scores), head circumference (centimeters or z scores), and length (centimeters or z scores) at 36 weeks' post menstrual age and at discharge.
4. Elevation of serum levels of aspartate aminotransferase (AST), alanine aminotransferase (ALT), and gamma glutamyl transferase (GGT) to more than two times the upper limits of normal levels for two weeks or more.

5. Improvement in liver inflammation is measured by the return of elevated levels of AST, ALT, and GGT to less than two times the upper limits of normal levels.
6. Incidence of central line-associated bloodstream infections (number of bloodstream infections per 1000 central venous catheter days) (bloodstream infection defined as a laboratory-confirmed bloodstream infection that is not secondary to an infection at another body site) [39, 40].
7. Length of hospital stay (in days).
8. Duration of central venous catheterization (in days).
9. Duration of parenteral nutrition (in days).
10. Duration of intravenous lipids (in days).
11. Need for liver transplantation in the first year of life (the criteria for liver transplant are end-stage liver disease with established intestinal failure-associated liver disease; at least 50 cm of functional small bowel in the absence of an ileocecal valve, or 30 cm with the valve; at least 50% of the estimated daily calorie requirement previously tolerated as enteral feeds together with weight gain) [1].
12. All-cause mortality during the first year of life [37].
13. Lipoid pneumonia as defined by clinical and radiologic features of pneumonia that follow aspiration or inhalation of animal fats or mineral or vegetable oil.

Search methods for identification of studies

An Information Specialist (CC) wrote search strategies in consultation with the clinical authors (Supplementary material 1).

Electronic searches

We last searched the following databases on 16 December 2024 without language, publication type, or publication status restrictions.

1. Cochrane Central Register of Controlled Trials (CENTRAL), in the Cochrane Library, Issue 12 of 12;
2. The Cochrane Database of Systematic Reviews (CDSR), in the Cochrane Library, Issue 12 of 12;
3. Ovid MEDLINE (MEDALL) (1946 to 13 December 2024);
4. Embase, OVID (1974 to 13 December 2024).

Searches were limited to randomized studies or systematic reviews using the August 2024 version of the Cochrane Neonatal filters (https://neonatal.cochrane.org/literature-search-filters-neonatal-reviews).

Searching other resources

We searched trial registration records using CENTRAL and by independent searches of the following databases on 16 December 2024:

1. US National Institutes of Health Ongoing Trials Register ClinicalTrials.gov (clinicaltrials.gov);
2. ICTRP—World Health Organization International Clinical Trials Registry Platform (https://trialsearch.who.int/Default.aspx).

We undertook reference checking and contact with study authors. The primary report of included studies was checked for post-publication amendment and/or concerns.

Data collection and analysis

We collected data following Cochrane methods [41, 42].

Had we identified studies by one of the authors of this review, two other review authors would have independently undertaken the following: screening and selection; data extraction; risk of bias assessment; and assessing certainty of evidence.

Selection of studies

We downloaded all titles and abstracts retrieved by electronic searches to reference management software and removed duplicates; screening was conducted using Covidence [43]. Had we identified non-English studies, we would have used an online translation service, or sought translation by a speaker conversant in the language.

Two review authors (MPr and KHu) independently assessed titles and abstracts retrieved by the literature search based on Criteria for considering studies for this review. The full text of references retained following the title/abstract assessment was assessed independently by two authors (MPr and KHu). At any stage of screening, conflicts were resolved by discussion and by input from a third review author (MPa).

We documented the reasons for excluding studies during the full-text review in Supplementary material 3. We provided any information we were able to obtain about ongoing studies in Supplementary material 5. When we identified studies with insufficient information to allow data extraction, we categorized them as awaiting classification, and documented all available characteristics in Supplementary material 4; if data becomes available, studies awaiting classification will be assessed for inclusion in an update of this review.

Where studies had multiple publications, we collated the reports of the same study, so that each study, rather than each report, was the unit of interest for the review, and such studies have a single identifier with multiple references.

Had we identified studies where only a subset of participants was eligible for inclusion, and if the data were not available in the study report, we would have contacted study authors to obtain individual participant data to determine inclusion into the review.

We recorded the selection process in sufficient detail to complete a PRISMA flow diagram [34, 35].

We analyzed the clinical outcomes noted in the Outcome measures section. While performing subgroup comparisons, the control groups were compared separately with the respective interventions. While performing comprehensive analysis, the comparison was made between one control group (inclusive of control groups of all comparisons) with one comprehensive intervention group (inclusive of all different interventions).

Data extraction and management

Two review authors (MPr and KHu) independently performed data extraction using specifically designed data forms for trial inclusion and exclusion, data extraction, and for requesting additional published information from authors of the original reports. We piloted the form on two included studies.

We extracted the following characteristics from each included study:

1. Administrative details: study author(s), published or unpublished, year of publication, year in which study was conducted, disclosures of conflicts, funding support;
2. Study characteristics: study registration, study design type, study setting, method of randomization, blinding, intervention, stratification, and whether the trial was single or multicenter;
3. Details regarding participants: number randomized, number analyzed, birthweight, gestational age, inclusion criteria, exclusion criteria;
4. Interventions: initiation, type, dose, duration and dose of lipids;
5. Outcomes as mentioned above in Outcome measures, and any other outcomes reported in the study.

Two study authors (MPr and KHu) compared the extracted data and resolved any differences by discussion, and by input from a third review author (MPa).

We contacted the study investigators/authors for clarifications when queries arose regarding the type of enteral lipid preparation, data inquiries, and to confirm that there was no overlap in the subject population in two studies by the same authors. We documented information about such contact in the Characteristics of included studies section [Supplementary material 2]. One review author (MPr) transferred study characteristics and risk of bias judgments into Cochrane's statistical software RevMan for data entry [44] and data into the analysis section. Two review authors (MPr and KHu) checked and verified the transferred study data in the analysis section.

Had we identified studies where only a subset of participants was eligible for inclusion, and if the data were not available in the original publication, we would have contacted study authors to obtain individual participant data to determine inclusion into the review.

Risk of bias assessment in included studies

Two review authors (MPr and KHu) independently assessed the risk of bias. We resolved any disagreements by discussion, and by input from the third review author (MPa). The risk of bias (low, high, or unclear) of all included trials was assessed using the Cochrane Risk of Bias Tool 1 (RoB1) [45], for the following domains:

1. Sequence generation (selection bias);
2. Allocation concealment (selection bias);
3. Blinding of participants and personnel (performance bias);
4. Blinding of outcome assessment (detection bias);
5. Incomplete outcome data (attrition bias);
6. Selective reporting (reporting bias);
7. Any other bias.

We used Cochrane Risk of Bias Tool 1 (RoB1) to assess risk of bias across all included trials. We examined sequence generation and allocation concealment to determine whether randomization methods were truly random and adequately protected against selection bias. Blinding of participants, personnel, and outcome assessors were assessed separately for relevant outcomes to identify risks of performance or detection bias. Completeness of outcome data was evaluated by reviewing attrition, exclusions, and whether missing data were balanced or related to outcomes. Selective reporting bias was assessed by comparing reported outcomes with prespecified protocols. We also considered other potential sources of bias, including design-related issues or early trial termination. All risk of bias assessments were categorized as low, high, or unclear risk and documented in the characteristics of included studies, the summary of findings table and the risk of bias table. The detailed methodology of the risk of bias for each domain is given in Risk of Bias domains (Supplementary material 10).

Had we identified cluster-randomized trials, we would have considered the following risks of bias as recommended in Chapter 23 [46] of the *Cochrane Handbook for Systematic Reviews of Interventions* [41, 47]:

1. Whether there was recruitment bias when participants were recruited after randomization of clusters;
2. Whether there was a possibility of chance baseline imbalance between the randomized groups;
3. Whether there were missing clusters from a trial;
4. Whether incorrect statistical analysis methods were applied;
5. Whether there was comparability with individual randomized trials.

Measures of treatment effect

We analyzed dichotomous data as risk ratios (RRs) with 95% confidence intervals (CIs), and continuous data as mean differences (MDs) or standardized mean differences (SMDs). We undertook meta-analyses when treatments, participants, and the underlying clinical questions were similar to facilitate a sensible pooling of data. When multiple trial arms were reported in a single trial, we included only the relevant arms.

If there was a statistically significant reduction in RD with 95% CIs, we planned to calculate the number needed to treat for an additional beneficial outcome (NNTB) or the number needed to treat for an additional harmful outcome (NNTH).

Unit of analysis issues

The unit of analysis was the infant participating in individually randomized trials. An infant was considered only once in the analysis.

Had we identified cluster-randomized trials for inclusion, we would have conducted analysis at the same level as the allocation, using a summary measurement from each cluster. Since this might reduce the power of the study, we would apply alternative statistical methods that allowed analysis at the level of the individual while accounting for the clustering in the data as recommended in Chapter 23 [46] of the *Cochrane Handbook for Systematic Reviews of Interventions* [41, 47].

In trials with multiple arms, we combined groups to create a single pairwise comparison to closely match the definitions given in Types of interventions. We did not exclude any groups or interventions in those studies.

Dealing with missing data

We conducted the analysis on an intention-to-treat (ITT) basis for all included outcomes. Whenever possible, we analyzed participants within their originally assigned treatment groups, regardless of the treatment they actually received, or outcomes with important missing or unclear data. We contacted the original investigators via email to request additional information. The details of the correspondence are included in the section, Supplementary material 9. If further details were required, we followed our standard approach of reaching out to trial authors to obtain any relevant missing data. In cases where no response was received after two attempts, we proceeded with the available data. In the case of missing data, the number of participants with missing data was described in the section, Supplementary material 4.

Had data been missing for dichotomous outcomes, we would have included participants with incomplete or missing data in the sensitivity analysis by imputing them according to the following scenarios.

1. Extreme-case analysis favoring the experimental intervention (best-worst case scenario): none of the dropouts/participants lost from the experimental arm, but all the dropouts/participants lost from the control arm experienced the outcome, including all randomized participants in the denominator;
2. Extreme-case analysis favoring the control (worst-best case scenario): all dropouts/participants lost from the experimental arm, but none from the control arm experienced the outcome, including all randomized participants in the denominator.

For continuous outcomes, we calculated missing standard deviations using reported P values or CIs [41]. If the data were assumed to be missing at random, we would analyze the data without imputing any missing values.

We discussed the implications of the missing data in the Discussion section of the review.

Reporting bias assessment

We obtained study protocols of all the included studies, where possible, and compared outcomes reported in the protocols to those reported in the included studies. If we suspected reporting bias, we would have attempted to contact the study authors asking them to provide missing outcome data. Where this was not possible, and the missing data were thought to introduce serious bias, we planned to explore the impact of including such studies in the overall assessment of results by sensitivity analysis.

We would have created a funnel plot to assess the small study and publication bias if there had been a sufficient number of studies (> 10) reporting the same outcome. If publication bias had been suggested by a significant asymmetry of the funnel plot, we would have incorporated that into our assessment of certainty of evidence. When our review included fewer than 10 eligible studies for meta-analysis, we noted our inability to rule out publication bias and included that in our assessment of evidence.

Synthesis methods

We conducted our statistical analysis using Review Manager Software [44], following the recommendations of Cochrane Neonatal. We analyzed dichotomous data as risk ratios (RRs) with 95% confidence intervals (CIs), while we analyzed continuous data as mean differences (MDs) or standardized mean differences (SMDs). We described clinical and methodological diversity narratively and summarized the information in tables.

We used a random-effects model for meta-analysis due to the following:

1. The heterogeneity of neonates, as both surgical and non-surgical groups were included;
2. The broad and multifactorial nature of PNALD inclusion criteria;
3. The variability in the type, dose, and duration of enteral lipid interventions.

To estimate treatment effects and assess heterogeneity, we inspected forest plots and quantified heterogeneity using the I^2 statistic. We classified heterogeneity as follows:

1. Not important (I^2 = 0% to 40%);
2. Moderate (I^2 = 30% to 60%);
3. Substantial (I^2 = 50% to 90%);
4. Considerable (I^2 > 75%).

We used forest plots to visually represent study data.

When meta-analysis was deemed inappropriate, we summarized data narratively from the trials following Chapter 12 [48] of the *Cochrane Handbook for Systematic Reviews of Interventions* [47]and the *Synthesis Without Meta-analysis (SWiM) reporting guidance* [49].

Investigation of heterogeneity and subgroup analysis

Had we found sufficient studies (greater than 10), we would have performed the following subgroup analyses for the primary outcomes to explore causes of heterogeneity:

1. Gestational age:
   a. Term 37 0/7 weeks or greater;
   b. Preterm 36 6/7 weeks or fewer;
2. Birthweight:
   a. Low birthweight (1501 to 2500 g);
   b. Very low birthweight (1000 to 1500 g);
   c. Extremely low birthweight (less than 1000 g);
3. Duration of supplementation: a. total duration of supplementation less than four weeks; b. total duration of supplementation four weeks or greater;
4. Lipid dosing (actual dose received daily; cumulative);
5. Based on the cause of requirement or dependence on parenteral nutrition:
   a. Surgical gastrointestinal conditions as a cause (e.g. stage 3 necrotizing enterocolitis, spontaneous intestinal perforation, gastroschisis, omphalocele, intestinal atresia, volvulus);
   b. Non-surgical gastrointestinal conditions as a cause (e.g. functional intestinal failure, intestinal dysmotility).

While performing comprehensive analysis, the comparison was made between one control group (inclusive of control groups of all comparisons) with one comprehensive intervention group (inclusive of different interventions).

Equity-related assessment

An equity-related assessment was not conducted.

Sensitivity analysis

Had we identified sufficient data for studies with low and high risk of bias, such as characteristics of bias, participant, outcomes, publication status, we would have explored sensitivity analyses. We made informal comparisons between the different ways of estimating the effect under different assumptions.

Certainty of the evidence assessment

We used the GRADE approach, as outlined in the GRADE Handbook [50], to assess the certainty of evidence of the following outcomes:

1. Prevention of PNALD during the hospital stay and in the first year of life (cholestasis) assessed with: serum bilirubin levels follow-up: range from two weeks to 17 weeks;
2. Resolution of PNALD during the hospital stay and in the first year of life (defined as a sustained reduction of elevated conjugated bilirubin levels to below 2 mg/dL or less than 20% of the total bilirubin concentration for two weeks or more, or as defined by the investigators);
3. Feeding intolerance assessed with clinical observation follow-up: range 14 days to 565 days;
4. Time taken to achieve full enteral feeds: assessed with clinical assessment;
5. Length of hospital stay;
6. Need for liver transplant in the first year of life;
7. All-cause mortality up to hospital discharge and the first year of life.

Two review authors (MPr and KHu) independently assessed the certainty of the evidence for each of the outcomes above. We considered evidence from randomized controlled trials as high certainty but downgraded the evidence by one level for serious (or two levels for very serious) limitations based upon the following: design (risk of bias), consistency across studies, directness of the evidence, precision of estimates, and presence of publication bias. We used the GRADEpro GDT Guideline Development Tool [51]to create a summary of findings table to report the certainty of the evidence for the following comparison:

1. Any enteral lipids compared to placebo or no treatment in infants at risk or with parenteral nutrition-associated liver disease (Summary of findings 1).

The GRADE approach results in an assessment of the certainty of a body of evidence as one of four grades:

1. High: we are very confident that the true effect lies close to that of the estimate of the effect.
2. Moderate: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.
3. Low: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.
4. Very low: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

We justified all decisions to downgrade the certainty of the evidence using footnotes, and we made comments to aid the reader's understanding of the review where necessary.

Consumer involvement

We did not involve consumers in this review due to limited resources. However, we used core outcome sets for the review's outcomes.

RESULTS

Description of studies

Results of the search

The searches for this review covered the period from 1946 to 16 December 2024. We searched three databases and two trial registries, as described in Search methods for identification of studies. Searches identified 12,038 references; after removing 5018 duplicates, 7020 were screened. We excluded 6970 based on the title abstract (98 by the Covidence RCT classifier; 6872 by authors), and reviewed 50 full texts or trial registry records. We included 11 studies (21 references); excluded 26 studies (26 references); classified one study (two references) as awaiting classification; and identified one ongoing study (one reference). Figure 1 shows the screening process.

Figure 1. Study flow diagram

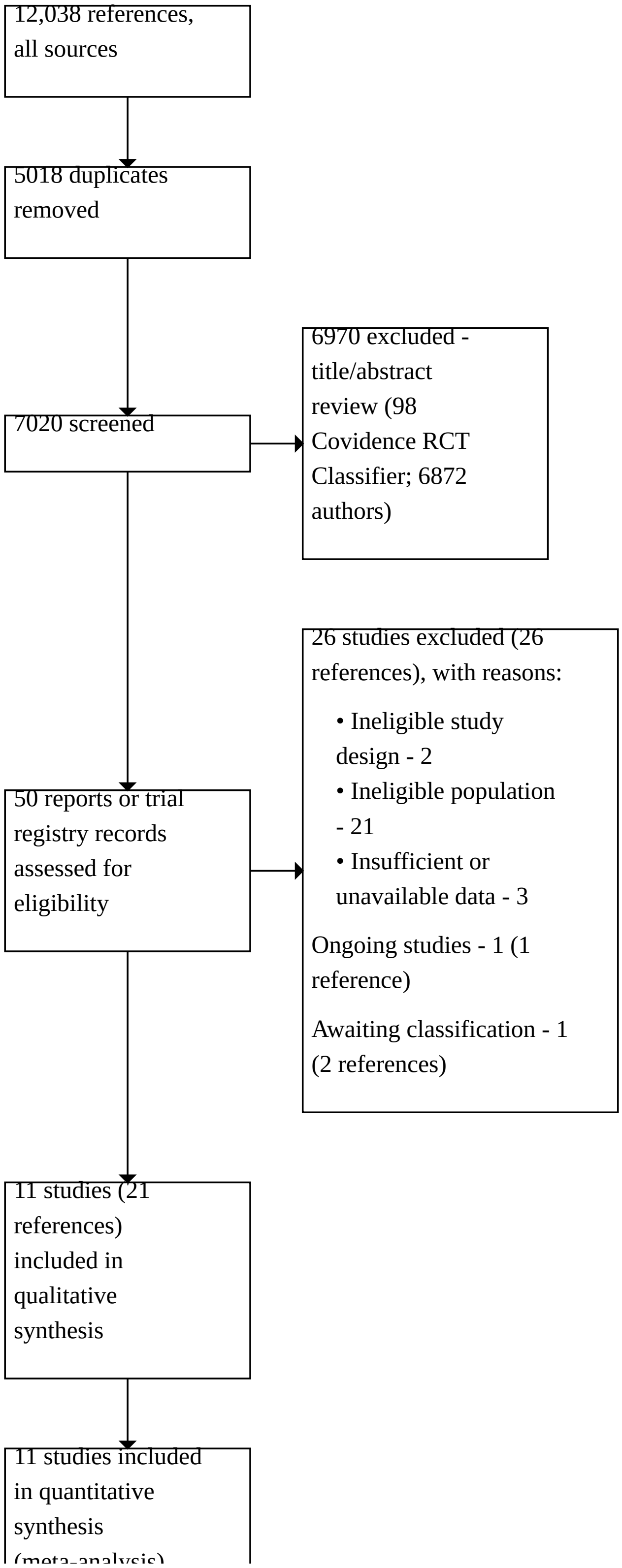


Figure 1. (Continued)

synthesis
(meta-analysis)

Included studies

Characteristics of studies

We included 11 studies involving 2192 participants that met the eligibility criteria. Studies were conducted in North America, South America, Europe, Asia, and Australasia. Table 1 provides an overview of the included studies and syntheses. Key study characteristics and additional details of the included studies are available in Supplementary material 2. Of the 11 studies, two (EMLFO Study [52]and ImNuT Study [53]) described outcomes of interest in five separate publications [54, 55, 56, 57, 58]. Seven studies involving eight reports were double-blind parallel-arm RCTs, and four studies were open-label RCTs described in five reports [54, 55, 59, 60, 61]. All studies except EMLFO-2 [61] were published as full peer-reviewed manuscripts. The study results of the EMLFO-2 Study [61] were obtained from clinicaltrials.gov.

Characteristics of participants

Since the Objectives of this review focused on infants who either had PNALD or were at risk of PNALD, the study population comprised of infants who were expected to be on parenteral nutrition for a significant amount of time. Details of the age and sex distribution of the participants are shown in Table 1 and Supplementary material 2. The study population included mainly preterm infants with and without gastrointestinal surgical conditions who were on parenteral nutrition and minimal enteral feeds. Male infants comprised 53% of the participants. Most of the infants had < 1500 g birthweight. The gestational age of the participants was between 24 and 34 weeks at birth. The surgical group of participants in the EMLFO study [54, 55, 62, 63] were less than two months old and had received an enterostomy for surgical gastrointestinal conditions.

Treatment format

In all the included studies, enteral lipids were administered at the bedside by mouth or through a feeding tube. Enteral lipids comprised various oils, including single oils and combinations of different oils, both in the intervention and control arms. Broadly, the intervention enteral lipids were categorized into two groups based on the type of enteral lipids:

1. Fish oil group:
   a. Fish oil alone [54, 55, 62, 63, 64, 65, 66], or
   b. A combination of fish oil with medium chain triglyceride [61].
2. Algal oil group:
   a. Algal oil alone [59, 67, 68, 69], or
   b. A combination of algal oil and fungal oil [56, 57, 58, 60, 70, 71].

In two studies [55, 59], the comparison group did not receive any placebo or control emulsion. In the included studies, algal oil and fish oil were used as a source of docosahexaenoic acid (DHA), and fungal oil as a source of arachidonic acid (ARA). In those studies that provided placebo or sham oil to the control group, it comprised either sunflower oil, safflower oil, olive oil, or soy oil.

The duration of the intervention between the studies was variable, with some adopting a fixed duration from two weeks to eight weeks or until a certain post-menstrual age, ranging from 36 weeks to 40 weeks. In studies with post-surgical participants, the intervention was administered until full feeds were achieved and cholestasis resolved [54, 55].

Outcomes reporting and assumptions required for synthesis

None of the 11 included studies considered either prevention or treatment of PNALD as their primary outcome. Three studies [60, 70, 71] that reported incidence of cholestasis were included in the prevention of PNALD analysis, while one study [55] that reported only the levels of conjugated bilirubin could not be included. In several instances, the data could not be included, since they were represented in units other than the mean (standard deviation), and the normality could not be assumed (time to full feeds [68, 70, 71], growth [55, 71], length of stay [64, 68, 69, 70, 71], and duration of parenteral nutrition [55, 69, 71]). The reports of blood culture-confirmed infection were considered under central line-associated bloodstream infections, and those studies which included suspected sepsis were excluded from the analysis [60, 68, 69, 70].

Contact with study authors

Author queries for five studies clarified key details: Robinson 2016 contained a numerical error [71]; NCT01674478 had not been published in a peer-reviewed journal [61]; Bernabe-Garcia 2019 and 2021 involved different study populations [68, 69]; and Baack 2016 provided clarification on the enteral lipid intervention [67].

Excluded studies

We excluded 26 studies after full-text review (see Supplementary material 3).

The most common reason for exclusion was ineligible study population (N = 21): studies involving infants not at risk for PNALD due to lack of parenteral nutrition exposure [72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92]. Two studies were excluded due to an ineligible design lacking appropriate controls [93, 94] and three due to incomplete data [95, 96, 97].

Studies awaiting classification

We found one study awaiting classification [98]. This study was classified as "Awaiting Classification" due to missing crucial details in the publication. We contacted the authors via email on 3 December 2024 and 27 December 2024, but are still awaiting a response. See Supplementary material 4.

Ongoing studies

We identified one ongoing study [99], an RCT in the USA, investigating early DHA/ARA supplementation in growth-restricted, very preterm infants. Infants born between 22 (0/7) and 32 (6/7) weeks of gestation receive(d) DHA/ARA supplementation orally within the first three weeks after birth. They are being compared to a control group receiving standard care. Outcomes include growth, serum metabolic profile, and cognitive development. See Supplementary material 5.

Risk of bias in included studies

We have included risk of bias assessments and the reasons for judgment across all ROB 1 domains for each study in Supplementary material 8 and Figure 2. The detailed methodology of the risk of bias for each domain is in Supplementary material 10.

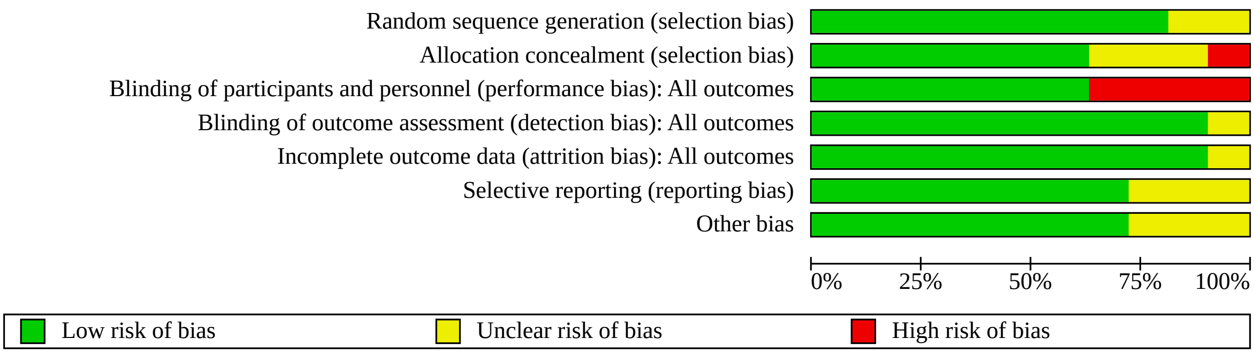
Figure 2. Risk of bias summary

	Random sequence generation (selection bias)	Allocation concealment (selection bias)	Blinding of participants and personnel (performance bias): All outcomes	Blinding of outcome assessment (detection bias): All outcomes	Incomplete outcome data (attrition bias): All outcomes	Selective reporting (reporting bias)	Other bias
Abou El Fadl 2021	+	+	−	?	+	?	?
Baack 2016	?	?	+	+	+	+	+
Bernabe-Garcia 2019	+	+	+	+	+	+	+
Bernabe-García 2021	+	+	+	+	+	?	+
Collins 2017	+	+	+	+	+	+	+
Frost 2021	+	+	+	+	+	+	+
Hellstrom 2021	+	−	−	+	+	+	+
NCT01674478	?	?	−	+	+	+	?
Robinson 2016	+	+	+	+	+	+	+
Wendel 2023	+	+	+	+	+	+	+
Yang 2014	+	?	−	+	?	?	?

Overall, the risk of selection bias was unclear to high. The process of randomization was satisfactory across all studies except two studies where no details of randomization were available to make a judgment, which was considered as unclear risk of bias [61, 67]. Inclusion of block randomization introduced a possibility of bias in allocation concealment in some studies [55, 60, 70]. Open-label study design raised concerns about performance bias in some of the included studies [55, 59, 60, 61]. Due to the objective definitions of the included outcomes, the detection bias was determined to be of low to unclear risk of bias in the included studies. The absence of outcomes for all randomized infants, and mismatch between outcomes in study protocols and published reports raised concerns about the risk of attrition and reporting bias [55, 59, 68]. In general, just under half of the included studies had a low risk of bias [58, 64, 69, 70, 71]. Due to concerns in at least one domain, four studies were considered at high risk for bias [55, 60, 61, 68].

A summary of the risk of bias by domain, taking into account all the included studies, is represented in the risk of bias graph (Figure 3).

Figure 3. Risk of bias graph



Synthesis of results

The absolute and relative effects for all critical and selected important outcomes, for comparisons of any enteral lipids versus no treatment or placebo, are summarized with the certainty of the evidence ratings of the effect estimates in Summary of findings 1.

Comparison of any fish oil versus no treatment or placebo

Critical outcomes

Prevention of PNALD during the hospital stay and in the first year of life (incidence of cholestasis)

No studies reported this outcome.

Resolution of PNALD during the hospital stay and in the first year of life

No studies reported this outcome.

Important outcomes

Feeding intolerance

One study reported this outcome (Yang 2014 [52, 54, 55, 62, 63, 65, 66]). None of the infants randomized to either fish oil enteral lipid or control were reported to have feed intolerance (Not estimable; 1 study, 36 participants; Analysis 1.1 in Supplementary material 6).

Time taken to achieve full enteral feeds during the hospital stay (time to full feeds)

One study reported the time taken to achieve full enteral feeds (Collins 2017 [64]). Infants randomized to enteral lipids had similar time to achieve full feeds compared to those who received a placebo (MD −1.10, 95% CI −2.98 to 0.78; I^2 not applicable; 1 study, 1273 participants; Analysis 1.2 in Supplementary material 6).

Adequacy of growth

One study was included in the analysis (Collins 2017). Infants at risk for PNALD randomized to enteral lipid supplementation did not show a difference in growth as measured by weight z-scores (MD 0.02, 95% CI −0.09 to 0.13; I^2 not applicable; 1 study, 1205 participants; Analysis 1.3), length z-scores (MD 0.75, 95% CI 0.64 to 0.86; I^2 not applicable; 1 study, 1205 participants; Analysis 1.4), and head circumference z-scores (MD 0.00, 95% CI −0.14 to 0.14; I^2 not applicable; 1 study, 1205 participants; Analysis 1.5 in Supplementary material 6).

Elevation of serum levels of aspartate aminotransferase (AST), alanine aminotransferase (ALT), and gamma glutamyl transferase (GGT)

No studies reported this outcome.

Improvement in liver inflammation measured by the return of elevated levels of AST, ALT, and GGT

No studies reported this outcome.

Incidence of central line-associated bloodstream infections

Two studies with 1270 infants were eligible for the meta-analysis (Collins 2017; Yang 2014). Enteral lipid supplementation in infants at risk for PNALD resulted in no difference in the risk of developing central line-associated bloodstream infections compared to controls (RR 0.89, 95% CI 0.50 to 1.59; I^2 = 0%; 2 studies, 1270 participants; Analysis 1.6 in Supplementary material 6).

Length of hospital stay (in days)

One study reported this outcome in the median and interquartile range (Collins 2017). Since normality could not be assumed, these data could not be utilized in this analysis.

Duration of central venous catheterization (in days)

No studies reported this outcome.

Duration of parenteral nutrition (in days)

One study reported duration of parenteral nutrition and was eligible for systematic review (Yang 2014). Infants randomized to enteral lipids had similar duration of parenteral nutrition compared to control (MD −3.00, 95% CI −12.89 to 6.89; I^2 not applicable; 1 study, 36 participants; Analysis 1.7 in Supplementary material 6).

Duration of intravenous lipids (in days)

One study reported the duration of intravenous lipids and was eligible for systematic review (Yang 2014). Infants randomized to enteral lipids had similar duration of intravenous lipids compared to controls (MD −5.00, 95% CI −11.43 to 1.43; I^2 not applicable; 1 study, 36 participants; Analysis 1.8 in in Supplementary material 6).

Need for liver transplantation in the first year of life

No studies reported this outcome.

All-cause mortality up to hospital discharge and the first year of life

Three studies were eligible for the meta-analysis (Collins 2017; NCT01674478 [61]; Yang 2014). Enteral lipid supplementation in infants at risk for PNALD did not result in a difference in all-cause mortality up to hospital discharge (RR 1.33, 95% CI 0.69 to 2.54; I^2 = 0%; 3 studies, 1328 participants; Analysis 1.9 in in Supplementary material 6).

Lipoid pneumonia

No studies reported this outcome.

Comparison of any plant-based oil versus no treatment or placebo

No studies reported this comparison.

Comparison of mixed oil emulsion versus no treatment or placebo

No studies reported this comparison.

Comparison of other enteral lipids (not defined above) versus no treatment or placebo

Eligible studies where any algal oil or fungal oil was used as an experimental lipid supplement were included here.

Critical outcomes

Prevention of PNALD disease during the hospital stay and in the first year of life (incidence of cholestasis)

Three studies that measured the incidence of cholestasis were included in the analysis (Frost 2021 [70]; Hellstrom 2021 [60]; Robinson 2016 [71]). One study measured the level of cholestasis (mg/dL) but not the incidence (Yang 2014); hence, this study was not included in the analysis. The biochemical measurements of serum bilirubin levels (mg/dL) ranged over a follow-up period of two weeks to 17 weeks. Participants randomized to enteral lipids had a similar risk of developing cholestasis compared to those who received a placebo (RR 0.65, 95% CI 0.13 to 3.14; I^2 = 0%; 3 studies, 265 participants; Analysis 2.1 in Supplementary material 6).

Resolution of PNALD during the hospital stay and in the first year of life

No studies reported this outcome.

Important outcomes

Feeding intolerance

Three studies measured this outcome by clinical observation over a follow-up period ranging from 14 days to 56 days (Baack 2016 [67]; Bernabe-Garcia 2019 [69]; Frost 2021). Two of the three studies assessed feed intolerance but did not report any intolerance (Baack 2016; Frost 2021). Infants randomized to enteral lipids had a similar risk of developing feeding intolerance compared to the controls (RR 0.14, 95% CI 0.02 to 1.12; I^2 not applicable; 3 studies, 200 participants; Analysis 2.2 in Supplementary material 6).

Time taken to achieve full enteral feeds during the hospital stay (time to full feeds)

Three studies reported this outcome in median and either range or interquartile range (Bernabe-Garcia 2019; Frost 2021; Robinson 2016). Since normality could not be assumed, these data could not be utilized in this analysis.

Adequacy of growth

One study was included in the analysis. Enteral lipid supplementation in infants at risk for PNALD did not result in a difference in growth as measured by: weight z-scores (MD −0.09, 95% CI −0.58 to 0.40; I^2 not applicable; 1 study, 97 participants; Analysis 2.3); length z-scores (MD 0.07, 95% CI −0.46 to 0.60; I^2 not applicable; 1 study, 97 participants; Analysis 2.4 in Supplementary material 6); and head circumference z-scores (MD −0.15, 95% CI −0.63 to 0.33; I^2 not applicable; 1 study, 97 participants; Analysis 2.5 in Supplementary material 6).

Elevation of serum levels of aspartate aminotransferase (AST), alanine aminotransferase (ALT), and gamma glutamyl transferase (GGT)

No studies reported this outcome.

Improvement in liver inflammation measured by the return of elevated levels of AST, ALT, and GGT

No studies reported this outcome.

Incidence of central line-associated bloodstream infections

Three studies with 208 infants were eligible for the meta-analysis (Baack 2016; Robinson 2016; Wendel 2023 [53, 56, 57, 58]). Enteral lipid supplementation in infants at risk for PNALD did not result in a difference in the risk of developing central line-associated bloodstream infections compared to controls (RR 1.10, 95% CI 0.53 to 2.25; I^2 = 0%; 3 studies, 208 participants; Analysis 2.6 in Supplementary material 6).

Length of hospital stay (in days)

Five studies reported this outcome (Abou El Fadl 2021 [59, 100]; Bernabe-Garcia 2019; Bernabe-García 2021 [68]; Frost 2021; Robinson 2016). However, four of those studies reported this outcome in the median and interquartile range. Since normality could not be assumed, these data could not be utilized in this analysis (Bernabe-Garcia 2019; Bernabe-García 2021; Frost 2021; Robinson 2016). One study reported length of stay and was eligible for the systematic review (Abou El Fadl 2021). The length of stay was shorter in the enteral lipid supplementation group compared to those who received a placebo (MD −3.44, 95% CI −6.20 to −0.68; I^2 not applicable; 1 study, 67 participants; Analysis 2.7 in Supplementary material 6).

Duration of central venous catheterization (in days)

No studies reported this outcome.

Duration of parenteral nutrition (in days)

Two studies reported this outcome with the median and interquartile range (Bernabe-Garcia 2019; Robinson 2016). Since normality could not be assumed, these data could not be utilized in this analysis.

Duration of intravenous lipids (in days)

Two studies reported this outcome with the median and interquartile range (Frost 2021; Robinson 2016). Since normality could not be assumed, these data could not be utilized in this analysis.

Need for liver transplantation in the first year of life

No studies reported this outcome.

All-cause mortality up to hospital discharge and the first year of life

Eight studies were eligible for the meta-analysis (Abou El Fadl 2021; Baack 2016; Bernabe-Garcia 2019; Bernabe-García 2021; Frost 2021; Hellstrom 2021; Robinson 2016; Wendel 2023). Enteral lipid supplementation in infants at risk for PNALD did not result in a difference in all-cause mortality up to hospital discharge (RR 1.01, 95% CI 0.62 to 1.65; I^2 = 0%; 8 studies, 872 participants; Analysis 2.8 in Supplementary material 6).

Lipoid pneumonia

No studies reported this outcome.

Comparison of one enteral lipid emulsion versus another enteral lipid emulsion

There were no eligible studies where any comparisons were made between one enteral lipid emulsion versus another.

Comparisons of any enteral lipids versus no treatment or placebo

Eleven studies were included in this comparison.

A summary of key results is in Summary of findings 1.

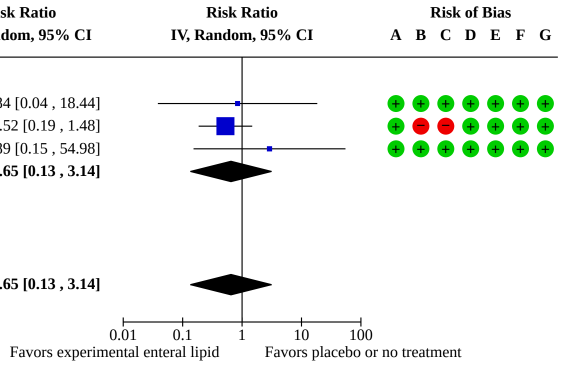
Critical outcomes

Prevention of PNALD during the hospital stay and in the first year of life (incidence of cholestasis)

Three studies that measured the incidence of cholestasis were included in the analysis (Frost 2021; Hellstrom 2021; Robinson 2016). One study measured the level of cholestasis (mg/dL) but not the incidence (Yang 2014); hence, this study was not included in the analysis. The biochemical measurements of serum bilirubin levels (mg/dL) ranged over a follow-up period of two weeks to 17 weeks. Enteral lipids may have little to no effect on the risk of developing PNALD during the hospital stay and in the first year of life. However, the evidence is very uncertain (RR 0.65, 95% CI 0.13 to 3.14; I^2 = 0%; 3 studies, 265 participants; very low-certainty evidence; Figure 4; Analysis 3.1). The certainty of the evidence was downgraded due to concerns about risk of bias in allocation and blinding, non-inclusion of high-risk infants (e.g. those with intestinal failure), and imprecision resulting from small sample sizes and wide confidence intervals.

Figure 4. Comparisons of any enteral lipids to placebo or no treatment in infants at risk or with parenteral nutrition-associated liver disease. 3.1 Prevention of parenteral nutrition-associated liver disease during the hospital stay and in the first year of life (Incidence of cholestasis)

Study or Subgroup	Experimental enteral lipid Events	Experimental enteral lipid Total	Placebo or no treatment Events	Placebo or no treatment Total	Weight	Risk Ratio IV, Random, 95% CI
3.1.1 Any Algal oil enteral lipid						
Frost 2021	1	24	0	6	9.1%	0.84 [0.04 , 18.44]
Hellstrom 2021	5	101	10	106	80.8%	0.52 [0.19 , 1.48]
Robinson 2016	2	18	0	10	10.1%	2.89 [0.15 , 54.98]
Subtotal (HKSJa)		**143**		**122**	**100.0%**	**0.65 [0.13 , 3.14]**
Total events:	8		10			
Test for overall effect: T = 1.18, df = 2 (P = 0.36)						
Heterogeneity: Tau² (REMLb) = 0.00; Chi² = 1.18, df = 2 (P = 0.55); I² = 0%						
Total (HKSJa)		**143**		**122**	**100.0%**	**0.65 [0.13 , 3.14]**
Total events:	8		10			
Test for overall effect: T = 1.18, df = 2 (P = 0.36)						
Test for subgroup differences: Not applicable						
Heterogeneity: Tau² (REMLb) = 0.00; Chi² = 1.18, df = 2 (P = 0.55); I² = 0%						



Footnotes

aCI calculated by Hartung-Knapp-Sidik-Jonkman (HKSJ) method.

bTau² calculated by Restricted Maximum-Likelihood method.

Risk of bias legend

(A) Random sequence generation (selection bias)
(B) Allocation concealment (selection bias)
(C) Blinding of participants and personnel (performance bias)
(D) Blinding of outcome assessment (detection bias)
(E) Incomplete outcome data (attrition bias)
(F) Selective reporting (reporting bias)
(G) Other bias

Resolution of PNALD during the hospital stay and in the first year of life

No studies reported this outcome.

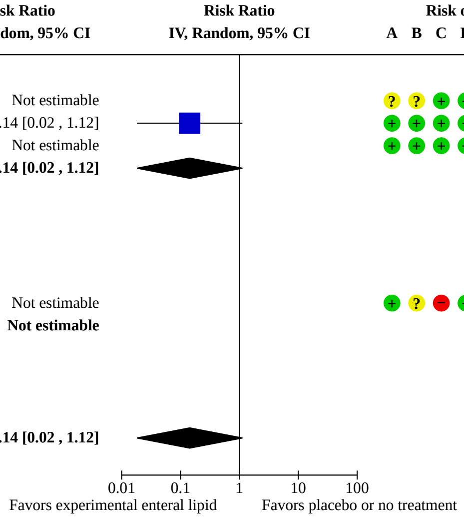
Important outcomes

Feeding intolerance

Four studies measured this outcome by clinical observation over a follow-up period ranging from 14 days to 56 days (Baack 2016; Bernabe-Garcia 2019; Frost 2021; Yang 2014). Three of the four studies assessed feed intolerance but did not report any intolerance (Baack 2016; Frost 2021; Yang 2014). Enteral lipids may have little to no effect on feeding intolerance, but the evidence is very uncertain (RR 0.14, 95% CI 0.02 to 1.12; I^2 not applicable; 4 studies, 236 participants; very low-certainty evidence; Figure 5; Analysis 3.2). The certainty of the evidence was downgraded due to inconsistent definitions of feed intolerance across studies, because infants requiring prolonged parenteral nutrition were not included, and because of imprecision stemming from a single small study with wide confidence intervals.

Figure 5. Comparisons of any enteral lipids to placebo or no treatment in infants at risk or with parenteral nutrition-associated liver disease. 3.2 Feeding intolerance

Study or Subgroup	Experimental enteral lipid Events	Experimental enteral lipid Total	Placebo or no treatment Events	Placebo or no treatment Total	Weight	Risk Ratio IV, Random, 95% CI	A	B	C	D	E	F	G
3.2.1 Any algal oil enteral lipid													
Baack 2016	0	31	0	29		Not estimable	?	?	+	+	+	+	+
Bernabe-Garcia 2019	1	55	7	55	100.0%	0.14 [0.02 , 1.12]	+	+	+	+	+	+	+
Frost 2021	0	24	0	6		Not estimable	+	+	+	+	+	+	+
Subtotal		**110**		**90**	**100.0%**	**0.14 [0.02 , 1.12]**							
Total events:	1		7										
Test for overall effect: Z = 1.85 (P = 0.06)													
Heterogeneity: Not applicable													
3.2.2 Any fish oil enteral lipid													
Yang 2014	0	18	0	18		Not estimable	+	?	−	+	?	?	?
Subtotal		**18**		**18**		**Not estimable**							
Total events:	0		0										
Test for overall effect: Not applicable													
Heterogeneity: Not applicable													
Total		**128**		**108**	**100.0%**	**0.14 [0.02 , 1.12]**							
Total events:	1		7										
Test for overall effect: Z = 1.85 (P = 0.06)													
Test for subgroup differences: Not applicable													
Heterogeneity: Not applicable													

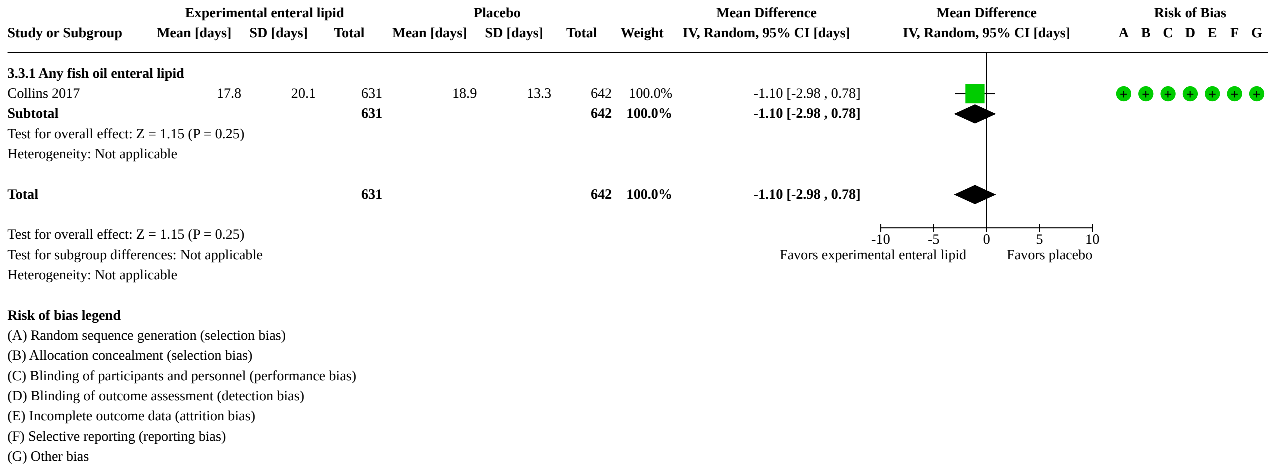


Risk of bias legend
(A) Random sequence generation (selection bias)
(B) Allocation concealment (selection bias)
(C) Blinding of participants and personnel (performance bias)
(D) Blinding of outcome assessment (detection bias)
(E) Incomplete outcome data (attrition bias)
(F) Selective reporting (reporting bias)
(G) Other bias

Time taken to achieve full enteral feeds during the hospital stay (time to full feeds)

Four studies reported the time taken to achieve full enteral feeds (Bernabe-García 2021; Collins 2017; Frost 2021; Robinson 2016). Three studies (Bernabe-García 2021; Frost 2021; Robinson 2016), reported time to full feeds in median and interquartile ranges; since normality could not be assumed, these data were not included in the review. These three studies that reported time to full feeds as median and interquartile ranges collectively suggested no clear effect of enteral lipid supplementation on the time required to achieve full enteral feeding. One study was eligible for the systematic review (Collins 2017). Enteral lipids may have little to no effect on time taken to achieve full enteral feeds during the hospital stay, but the evidence is very uncertain. Infants randomized to enteral lipids had a similar time to achieve full feeds compared to those who received a placebo (MD −1.10, 95% CI −2.98 to 0.78; I^2 not applicable; 1 study, 1273 participants; very low-certainty evidence; Figure 6; Analysis 3.3). The certainty of the evidence was downgraded due to indirectness related to a study population that did not require prolonged parenteral nutrition, substantial imprecision from a single study with wide confidence intervals, and concerns for publication bias given the lack of outcome reporting in other studies.

Figure 6. Comparisons of any enteral lipids to placebo or no treatment in infants at risk or with parenteral nutrition associated liver disease. 3.3 Time taken to achieve full enteral feeds (in days) during the hospital stay



Adequacy of growth

Two studies were included in the analysis (Collins 2017; Wendel 2023). Enteral lipid supplementation in infants at risk for PNALD did not result in a difference in growth as measured by: weight z-scores (MD 0.01, 95% CI −0.28 to 0.31; I^2 = 0%; 2 studies, 1302 participants; not specified; Figure 7; Analysis 3.4); length z-scores (MD 0.46, 95% CI −3.81 to 4.73; I^2 = 83%; 2 studies, 1302 participants; not specified; Figure 8; Analysis 3.5); and head circumference z-scores (MD 0.00, 95% CI −0.17 to 0.18; I^2 = 0%; 2 studies, 1302 participants; not specified; Figure 9; Analysis 3.6).

Figure 7. Comparisons of any enteral lipids to placebo or no treatment in infants at risk or with parenteral nutrition-associated liver disease. 3.4 Adequacy of growth - weight (z-scores)

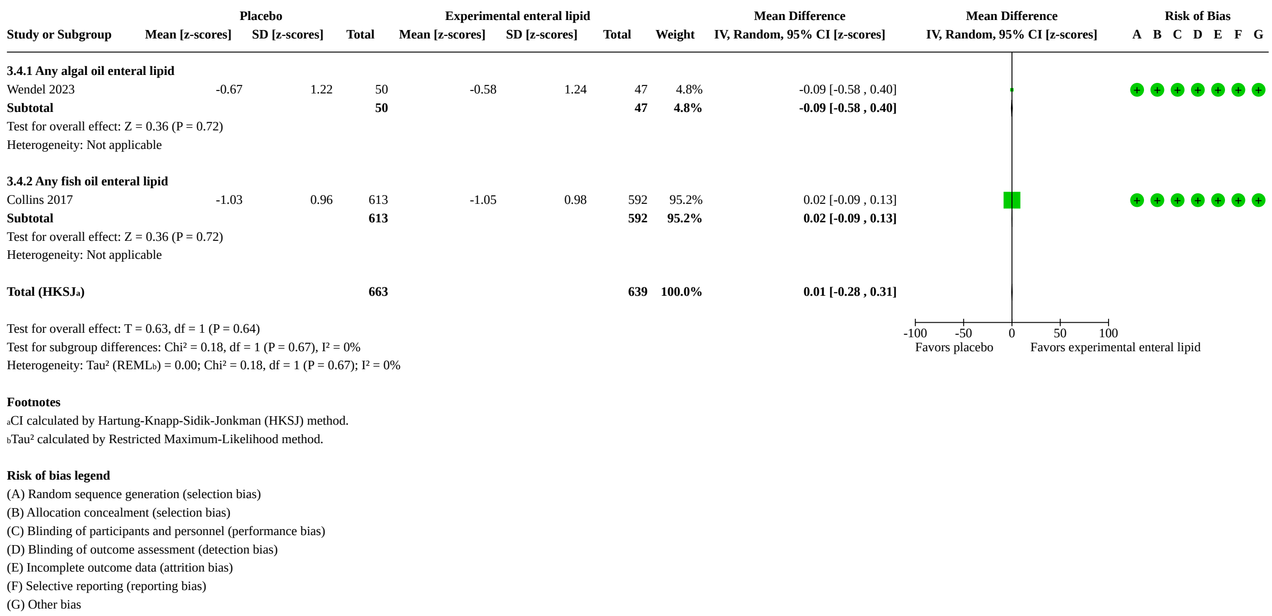


Figure 8. Comparisons of any enteral lipids to placebo or no treatment in infants at risk or with parenteral nutrition-associated liver disease. 3.5 Adequacy of growth - length (z-scores)

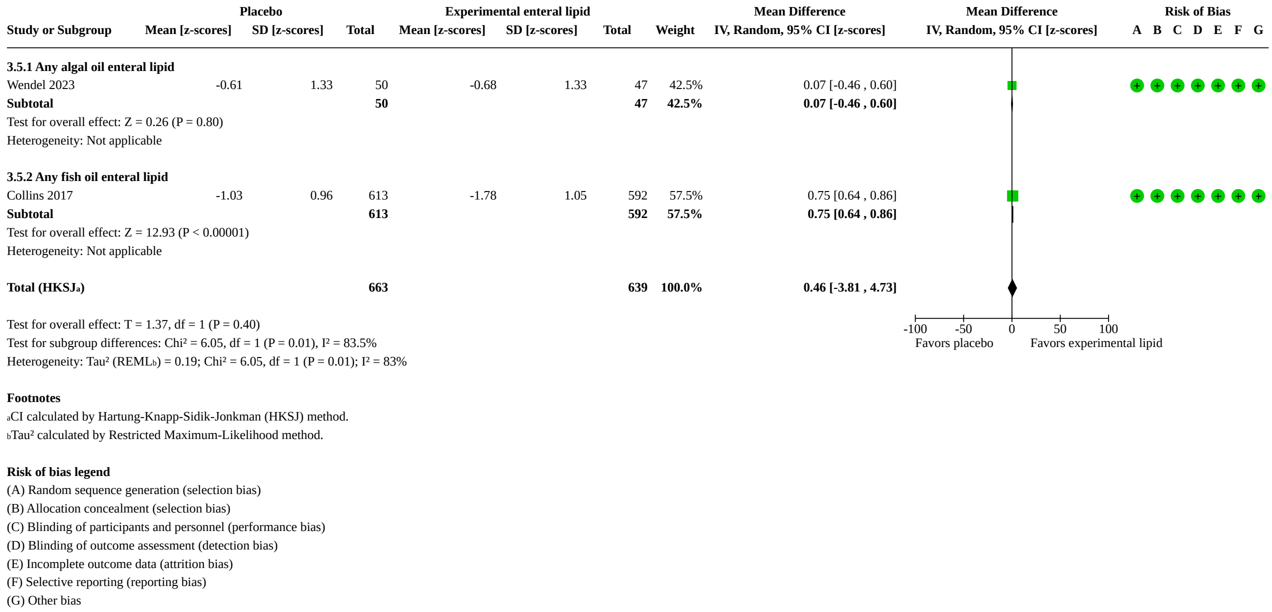
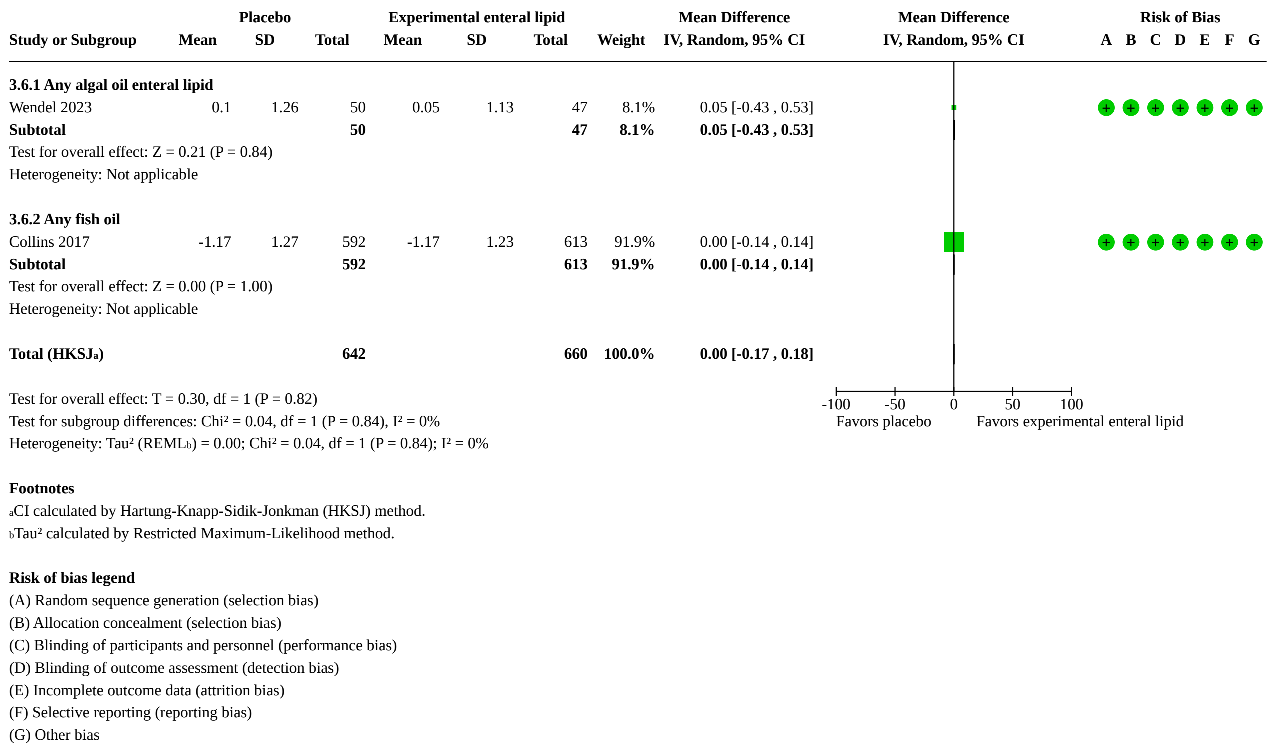


Figure 9. Comparisons of any enteral lipids to placebo or no treatment in infants at risk or with parenteral nutrition associated liver disease. 3.6 Adequacy of growth - head circumference (z-scores)



Elevation of serum levels of aspartate aminotransferase (AST), alanine aminotransferase (ALT), and gamma glutamyl transferase (GGT)

No studies reported this outcome.

Improvement in liver inflammation measured by the return of elevated levels of AST, ALT, and GGT

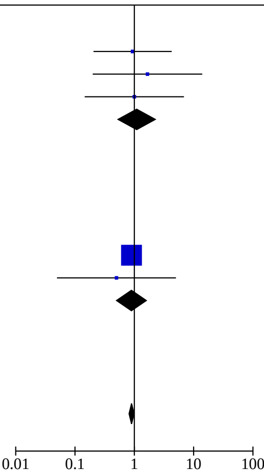
No studies reported this outcome.

Incidence of central line-associated bloodstream infections

Five studies with 1478 infants were eligible for the meta-analysis (Baack 2016; Collins 2017; Robinson 2016; Wendel 2023; Yang 2014). Enteral lipid supplementation in infants at risk for PNALD did not result in a difference in the risk of developing central line-associated bloodstream infections compared to controls (RR 0.90, 95% CI 0.82 to 0.99; I^2 = 0%; 5 studies, 1478 participants; not specified; Figure 10; Analysis 3.7).

Figure 10. Comparisons of any enteral lipids to placebo or no treatment in infants at risk or with parenteral nutrition-associated liver disease. Outcome 3.7 Incidence of central line-associated bloodstream infection

Study or Subgroup	Experimental enteral lipid Events	Experimental enteral lipid Total	Placebo or no treatment Events	Placebo or no treatment Total	Weight	Risk Ratio IV, Random, 95% CI	Risk of Bias A B C D E F G
3.7.1 Any algal oil enteral lipid							
Baack 2016	3	31	3	29	1.4%	0.94 [0.20 , 4.27]	? ? + + + + +
Robinson 2016	3	18	1	10	0.7%	1.67 [0.20 , 13.98]	+ + + + + + +
Wendel 2023	2	60	2	60	0.8%	1.00 [0.15 , 6.87]	+ + + + + + +
Subtotal (HKSJ[a])		**109**		**99**	**2.9%**	**1.10 [0.53 , 2.25]**	
Total events:	8		6				
Test for overall effect: T = 0.54, df = 2 (P = 0.64)							
Heterogeneity: Tau² (REML[b]) = 0.00; Chi² = 0.20, df = 2 (P = 0.90); I² = 0%							
3.7.2 Any fish oil enteral lipid							
Collins 2017	160	611	182	623	96.5%	0.90 [0.75 , 1.07]	+ + + + + + +
Yang 2014	1	18	2	18	0.6%	0.50 [0.05 , 5.04]	+ ? − + ? ? ?
Subtotal (HKSJ[a])		**629**		**641**	**97.1%**	**0.89 [0.50 , 1.59]**	
Total events:	161		184				
Test for overall effect: T = 2.49, df = 1 (P = 0.24)							
Heterogeneity: Tau² (REML[b]) = 0.00; Chi² = 0.24, df = 1 (P = 0.62); I² = 0%							
Total (HKSJ[a])		**738**		**740**	**100.0%**	**0.90 [0.82 , 0.99]**	
Total events:	169		190				
Test for overall effect: T = 3.08, df = 4 (P = 0.04)							
Test for subgroup differences: Chi² = 1.38, df = 1 (P = 0.24), I² = 27.3%							
Heterogeneity: Tau² (REML[b]) = 0.00; Chi² = 0.59, df = 4 (P = 0.96); I² = 0%							



Footnotes

[a] CI calculated by Hartung-Knapp-Sidik-Jonkman (HKSJ) method.

[b] Tau² calculated by Restricted Maximum-Likelihood method.

Risk of bias legend

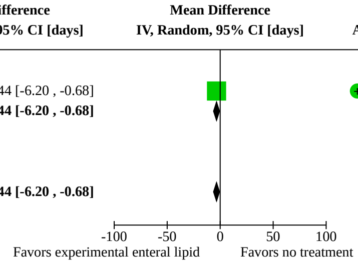
(A) Random sequence generation (selection bias)
(B) Allocation concealment (selection bias)
(C) Blinding of participants and personnel (performance bias)
(D) Blinding of outcome assessment (detection bias)
(E) Incomplete outcome data (attrition bias)
(F) Selective reporting (reporting bias)
(G) Other bias

Length of hospital stay (in days)

Six studies reported this outcome (Abou El Fadl 2021; Bernabe-Garcia 2019; Bernabe-García 2021; Collins 2017; Frost 2021; Robinson 2016). However, five of those studies reported this outcome in the median and interquartile range. Since normality could not be assumed, these data could not be utilized in this analysis (Bernabe-Garcia 2019; Bernabe-García 2021; Collins 2017; Frost 2021; Robinson 2016). One study that reported length of stay was eligible for systematic review (Abou El Fadl 2021). Enteral lipid supplementation may decrease the length of stay, but the evidence is very uncertain (MD −3.44, 95% CI −6.20 to −0.68; I^2 not applicable; 1 study, 67 participants; very low-certainty evidence; Figure 11; Analysis 3.8). The certainty of the evidence was downgraded due to risk of bias from performance and detection issues, because high-risk infants (e.g. those with intestinal failure) were not included, imprecision from a single small study with wide confidence intervals, and concerns for publication bias. Though results from one study suggested that enteral lipid supplementation may decrease the length of stay, the other five studies that reported length of stay in the median and interquartile range collectively showed no consistent reduction in hospitalization. Hence, the true effect may be substantially different from the estimate of the effect.

Figure 11. Comparisons of any enteral lipids to placebo or no treatment in infants at risk or with parenteral nutrition-associated liver disease. 3.8 Length of hospital stay

	Experimental enteral lipid			No treatment				Mean Difference	Risk of Bias
Study or Subgroup	Mean [days]	SD [days]	Total	Mean [days]	SD [days]	Total	Weight	IV, Random, 95% CI [days]	A B C D E F G
3.8.1 Any algal oil enteral lipid									
Abou El Fadl 2021	21.63	6.67	33	25.07	4.67	34	100.0%	-3.44 [-6.20 , -0.68]	+ + − ? + ? ?
Subtotal			**33**			**34**	**100.0%**	**-3.44 [-6.20 , -0.68]**	
Test for overall effect: Z = 2.44 (P = 0.01)									
Heterogeneity: Not applicable									
Total			**33**			**34**	**100.0%**	**-3.44 [-6.20 , -0.68]**	



Test for overall effect: Z = 2.44 (P = 0.01)
Test for subgroup differences: Not applicable
Heterogeneity: Not applicable

Risk of bias legend
(A) Random sequence generation (selection bias)
(B) Allocation concealment (selection bias)
(C) Blinding of participants and personnel (performance bias)
(D) Blinding of outcome assessment (detection bias)
(E) Incomplete outcome data (attrition bias)
(F) Selective reporting (reporting bias)
(G) Other bias

Duration of central venous catheterization (in days)

No studies reported this outcome.

Duration of parenteral nutrition (in days)

Three studies reported this outcome; however, two of those studies reported this outcome in the median and interquartile range (Bernabe-Garcia 2019; Robinson 2016). Since normality could not be assumed, these data could not be utilized in this analysis. One study reported duration of parenteral nutrition and was eligible for systematic review (Yang 2014). Infants randomized to enteral lipids had similar duration of parenteral nutrition compared to control (MD −3.00, 95% CI −12.89 to 6.89; I^2 not applicable; 1 study, 36 participants; Analysis 3.9 in Supplementary material 6).

Duration of intravenous lipids (in days)

Three studies reported this outcome (Frost 2021; Robinson 2016; Yang 2014). However, two of those studies reported this outcome in the median and interquartile range (Frost 2021; Robinson 2016). Since normality could not be assumed, these data could not be utilized in this analysis. One study that reported the duration of intravenous lipids was eligible for the systematic review (Yang 2014). Infants randomized to enteral lipids had similar duration of intravenous lipids compared to controls (MD −5.00, 95% CI −11.43 to 1.43; I^2 not applicable; 1 study, 36 participants; Analysis 3.10 in Supplementary material 6).

Need for liver transplantation in the first year of life

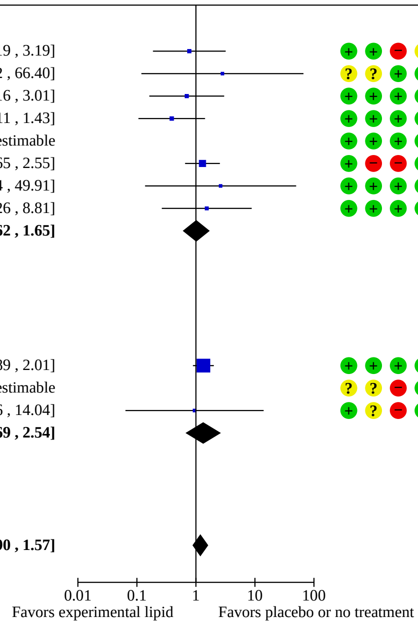
No studies reported this outcome.

All-cause mortality up to hospital discharge and the first year of life

Eleven studies were eligible for the meta-analysis (Abou El Fadl 2021; Baack 2016; Bernabe-Garcia 2019; Bernabe-García 2021; Collins 2017; Frost 2021; Hellstrom 2021; NCT01674478; Robinson 2016; Wendel 2023; Yang 2014). Enteral lipids may have little to no effect on mortality up to discharge or during the first year of life, but the evidence is very uncertain (RR 1.19, 95% CI 0.90 to 1.57; I^2 = 0%; 11 studies, 2200 participants; very low-certainty evidence; Figure 12; Analysis 3.11). The certainty of the evidence was downgraded due to risk of bias related to randomization and allocation concealment, inconsistency arising from wide variation in the type, dose, and duration of enteral lipid interventions, and indirectness because infants requiring prolonged parenteral nutrition were not included in the studies.

Figure 12. Comparisons of any enteral lipids to placebo or no treatment in infants at risk or with parenteral nutrition-associated liver disease.3.11 All-cause mortality up to hospital discharge and first year of life

Study or Subgroup	Experimental enteral lipid Events	Total	Placebo or no treatment Events	Total	Weight	Risk Ratio IV, Random, 95% CI	A	B	C	D	E	F	G
3.11.1 Any algal oil enteral lipid													
Abou El Fadl 2021	3	33	4	34	4.8%	0.77 [0.19 , 3.19]	+	+	-	?	+	?	?
Baack 2016	1	31	0	29	1.0%	2.81 [0.12 , 66.40]	?	?	+	+	+	+	+
Bernabe-Garcia 2019	3	74	4	69	4.5%	0.70 [0.16 , 3.01]	+	+	+	+	+	+	+
Bernabe-García 2021	3	105	8	109	5.7%	0.39 [0.11 , 1.43]	+	+	+	+	+	?	+
Frost 2021	0	24	0	6		Not estimable	+	+	+	+	+	+	+
Hellstrom 2021	16	101	13	106	20.8%	1.29 [0.65 , 2.55]	+	-	-	+	+	+	+
Robinson 2016	2	20	0	10	1.1%	2.62 [0.14 , 49.91]	+	+	+	+	+	+	+
Wendel 2023	3	60	2	61	3.1%	1.52 [0.26 , 8.81]	+	+	+	+	+	+	+
Subtotal (HKSJ[a])		**448**		**424**	**40.9%**	**1.01 [0.62 , 1.65]**							
Total events:	31		31										
Test for overall effect: T = 0.06, df = 6 (P = 0.95)													
Heterogeneity: Tau² (REML[b]) = 0.00; Chi² = 3.97, df = 6 (P = 0.68); I² = 0%													
3.11.2 Any fish oil enteral lipid													
Collins 2017	50	631	38	642	57.8%	1.34 [0.89 , 2.01]	+	+	+	+	+	+	+
NCT01674478	0	9	0	9		Not estimable	?	?	-	+	+	+	?
Yang 2014	1	19	1	18	1.3%	0.95 [0.06 , 14.04]	+	?	-	+	?	?	?
Subtotal (HKSJ[a])		**659**		**669**	**59.1%**	**1.33 [0.69 , 2.54]**							
Total events:	51		39										
Test for overall effect: T = 5.56, df = 1 (P = 0.11)													
Heterogeneity: Tau² (REML[b]) = 0.00; Chi² = 0.06, df = 1 (P = 0.80); I² = 0%													
Total (HKSJ[a])		**1107**		**1093**	**100.0%**	**1.19 [0.90 , 1.57]**							
Total events:	82		70										
Test for overall effect: T = 1.42, df = 8 (P = 0.19)													
Test for subgroup differences: Chi² = 1.72, df = 1 (P = 0.19), I² = 41.9%													
Heterogeneity: Tau² (REML[b]) = 0.00; Chi² = 4.75, df = 8 (P = 0.78); I² = 0%													



Footnotes

[a]CI calculated by Hartung-Knapp-Sidik-Jonkman (HKSJ) method.

[b]Tau² calculated by Restricted Maximum-Likelihood method.

Risk of bias legend

(A) Random sequence generation (selection bias)
(B) Allocation concealment (selection bias)
(C) Blinding of participants and personnel (performance bias)
(D) Blinding of outcome assessment (detection bias)
(E) Incomplete outcome data (attrition bias)
(F) Selective reporting (reporting bias)
(G) Other bias

Lipoid pneumonia

No studies reported this outcome.

Narrative summary of evidence when meta-analysis was not possible

Since normality could not be assumed (time to full feeds [68, 70, 71], growth [55, 71], length of stay [64, 68, 69, 70, 71], duration of parenteral nutrition [55, 69, 70, 71]), we summarized data from the trials narratively following Chapter 12 [48] of the *Cochrane Handbook for Systematic Reviews of Interventions* [47] and the *Synthesis Without Meta-analysis (SWiM) reporting guidance* [49].

Time taken to achieve full enteral feeds during the hospital stay (time to full feeds)

Across the three studies reporting time to achieve full enteral feeds, no statistically significant differences were observed between infants receiving enteral lipid supplementation and those receiving placebo. Median time to achieve full enteral feeds was not different between enteral lipid and placebo groups in Bernabe-García 2021 (15 vs 17 days; P = 0.161), Frost 2021 (10 vs 13 days; P = 0.34), or Robinson 2016 (24 vs 18 days; P = 0.06). Overall, these studies collectively suggest no clear effect of enteral lipid supplementation on the time required to achieve full enteral feeding.

Length of hospital stay (in days)

Across five studies reporting hospital length of stay, enteral lipid supplementation did not demonstrate any plausible effect. Median lengths of stay were similar between enteral lipid and placebo groups in Frost 2021 (58.5 vs 47.5 days; P = 0.2), Bernabe-García 2021 (45 vs 46 days; P = 0.88), Robinson 2016 (107 vs 111 days; P = 0.65), Bernabe-Garcia 2019 (both 52 days; P = 0.45), and Collins 2017 (91 vs 87 days; P = 0.09). Collectively, these findings show no consistent reduction in hospitalization duration with enteral lipid administration.

Adequacy of growth

Across the two studies reporting growth outcomes,, Robinson 2016 reported discharge weight comparisons, while Yang 2014 reported daily weight gain. Robinson 2016 showed no significant difference in discharge weight, with median weights of 3003 g vs 3378 g (P = 0.58) between enteral lipid and placebo groups, respectively. In contrast, Yang 2014 found higher daily weight gain among infants receiving enteral lipids: 27 ± 11 g/day vs 20 ± 9 g/day in controls, (P < 0.05). Overall, evidence suggests a possible improvement in short-term weight gain with enteral lipids, though findings are not consistent across studies.

Duration of parenteral nutrition (in days)

Across the two studies reporting duration of parenteral nutrition, there was no evidence of plausible differences observed between groups. In Bernabe-Garcia 2019, the median parenteral nutrition days were similar in the enteral lipid and placebo groups (14 [3–43] vs 16 [4–50] days; P = 0.262). Likewise, the mean (SD) parenteral nutrition days between enteral lipid and placebo groups were similar in Yang 2014: 10 (13) vs. 13 (17) days; P > 0.05, respectively. Overall, enteral lipid supplementation did not reduce the duration of parenteral nutrition.

Reporting biases

We evaluated the presence of reporting bias by examining discrepancies between the stated primary and secondary outcomes and those actually reported. When study protocols were accessible, we compared them with the corresponding published articles to assess the potential for reporting bias. We also recorded studies that involved interventions in an infant population that met eligibility criteria but did not report any of the primary or secondary outcomes listed in the Characteristics of included studies *(*Supplementary material 2) and risk of bias table (Supplementary material 8). Inconsistency between the published protocol and the published report (Abou El Fadl 2021; Bernabe-García 2021) and the absence of outcomes for all the reported infants (Yang 2014) raised concerns about the high risk of reporting bias in three studies.

Due to the limited number of studies eligible for meta-analysis for the majority of the outcomes, our capacity to detect publication bias was substantially reduced. As a result, we acknowledged that we could not rule out the possibility of publication bias or the influence of small study effects.

DISCUSSION

Summary of main results

The results are summarized in Summary of findings 1.

Eleven studies (2192 infants) investigated enteral lipid supplements in infants who were at risk of PNALD. Except for one study (Collins 2017), the other 10 studies were small-to-moderately sized. All except two studies were conducted in healthy preterm infants, with two studies (NCT01674478; Yang 2014) including infants with surgical conditions. The review covered a wide variety of enteral lipids, which were grouped under two broad categories: algal or a combination of algal and fungal oil, and fish oil or its combination. In the eligible studies, these enteral lipids were primarily studied for their docosahexaenoic acid content, either in algal oil, fish oil, or a combination of docosahexaenoic acid and arachidonic acid.

There is insufficient evidence from RCTs on enteral lipid supplementation in infants to suggest or refute a clinically meaningful reduction in the incidence of PNALD. The available evidence suggests that enteral lipids in infants who are at risk for PNALD may result in little to no effect on prevention of PNALD, feeding intolerance, time to achieve full enteral feeds, or mortality, but the evidence is very uncertain. Enteral lipids may be associated with a decrease in the length of hospital stay, but the evidence is very uncertain.

None of the studies examined the resolution of PNALD during the hospital stay and in the first year of life, or the need for liver transplantation, making a meta-analysis unfeasible.

Limitations of the evidence included in the review

The protocol was designed to evaluate the impact of enteral lipids on the prevention and treatment of PNALD [36]. However, we underestimated the complexities of enteral lipids when used as an intervention. The heterogeneity of enteral lipid formulations used and the limited number of studies hindered our ability to effectively assess their individual effects**.** Additionally, the scarcity of studies restricted our ability to perform subgroup analyses. After screening eligible studies for reliability, one study remains under classification pending further information from the author. However, its inclusion is unlikely to change the overall findings or conclusions.

The data from studies included in this review were not able to address the important outcomes outlined in the protocol. Though a few studies included the incidence of cholestasis as one of their outcomes, this outcome was not described as the prevention of PNALD. None of the studies evaluated the treatment or resolution of PNALD as their outcome.

Most of the infants included in this review were not at high risk of PNALD. They were preterm infants and a very small portion (2.5%) of the population were infants with surgical gastrointestinal conditions. Infants with surgical gastrointestinal conditions are at a higher risk of PNALD due to prolonged need for parenteral nutrition and exposure to inflammation originating from the disease and the surgical interventions. Therefore, the results of these analyses might not be generalizable to infants with surgical gastrointestinal conditions.

The studies included in this meta-analysis (n = 265) reported a baseline PNALD rate of 8% in control groups and a risk ratio of 0.65. Given this low event rate, detecting a clinically meaningful effect is statistically challenging. More than 1000 infants would be required to demonstrate a 50% relative reduction in PNALD with adequate power (α = 0.05, 80% power). Thus, this meta-analysis was grossly underpowered to demonstrate a plausible effect with enteral lipids compared to placebo or no treatment in the prevention of PNALD as measured by the incidence of cholestasis.

The certainty of evidence was downgraded across outcomes due to multiple methodological concerns. Key limitations included risk of bias (issues with allocation, blinding, and performance/detection bias), indirectness arising from the absence of high-risk infants such as those with intestinal failure or prolonged parenteral nutrition, inconsistency in how outcomes such as feed intolerance were defined, and imprecision related to small sample sizes and wide confidence intervals. In some cases, lack of outcome reporting across studies also raised concerns about publication bias.

The narrative summary of evidence that could not be included in the meta-analysis because of the assumption of non-normality did not account for differences in the relative sizes of the studies and no statistical tests were applied in the context of summarizing effect estimates.

The overall lack of benefits from the use of enteral lipids in infants found in this review could be due to multiple reasons. The heterogeneity arising from the wide variation of experimental lipids made the comparisons challenging. Often, the vehicle for the experimental lipid was also another lipid, which was also used as a comparative lipid. Of the 11 studies, eight studies had some form of enteral lipid formulation given to the control population as a sham intervention. In only three studies comprising 14% of the study population, the control group did not receive any intervention. The administration of some form of enteral lipid to the control population could potentially have diminished the effects of interventional enteral lipids.

Limitations of the review processes

We performed the review according to standard Cochrane methodological recommendations to minimize bias. We followed Cochrane guidelines systematically during the literature search, data collection, assessment of risk of bias, rating of certainty of evidence and interpretation of data. The study selection, data extraction, and quality assessment were performed in duplicate. Disagreements were resolved by discussion among the authors.

Agreements and disagreements with other studies or reviews

As far as we are aware, this is the first systematic review and meta-analysis of the use of enteral lipid formulations in the prevention and treatment of PNALD. There are two recent meta-analyses which studied the effect of enteral long-chain polyunsaturated fatty acids on necrotizing enterocolitis (NEC) and retinopathy of prematurity (ROP) [101, 102].

In the meta-analyses by Alshaik and colleagues, 15 RCTs included 3963 infants [101]. The objective of this meta-analysis was to investigate the efficacy of docosahexaenoic acid supplementation either alone or in combination with arachidonic acid on the incidence of NEC in preterm infants. This analysis found that "supplementation with docosahexaenoic acid alone increased necrotizing enterocolitis (2620 infants; RR: 1.56; 95% CI: 1.02, 2.39)." This study also reported a "significant reduction in necrotizing enterocolitis when arachidonic acid was supplemented with docosahexaenoic acid (aRR 0.42; 95% CI: 0.21, 0.88)." While this meta-analysis also examined enteral lipids, it primarily focused on feeds supplemented with docosahexaenoic acid and arachidonic acid. The study by Alshaik and colleagues included fatty acids delivered either as oil added at the bedside or integrated into the formula during commercial manufacturing [101]. However, their analysis did not assess the prevention or treatment of PNALD. In contrast, our Cochrane review evaluated enteral lipids more broadly, without specifying content, and specifically examined prevention and treatment of PNALD as critical outcomes.

In the meta-analyses by Diggikar and colleagues, nine RCTs included 2482 infants [102]. The aim of this meta-analysis was to assess the effect of long-chain polyunsaturated fatty acids on retinopathy of prematurity in preterm infants. This analysis deduced that supplementation of long-chain polyunsaturated fatty acids did not reduce the incidence of any retinopathy of prematurity (RR 0.95, 95% CI 0.73 to 1.12, 6 studies, 1177 infants), or severe retinopathy of prematurity (RR 0.71, 95% CI 0.50 to 1.01; 5 studies, 1822 infants). Similar to the meta-analysis by Alshaik and colleagues [101], this meta-analysis differed from ours in that they included long-chain polyunsaturated fatty acid supplementation either in oil form with the feeds at the bedside or integrated into the formula during commercial manufacturing. Also, in this review by Diggikar and colleagues, the primary outcome of either prevention or treatment of PNALD was not studied [102].

AUTHORS' CONCLUSIONS

Implications for practice

There is insufficient evidence from randomized controlled trials on enteral lipid supplementation in infants to suggest or refute a clinically meaningful effect on the prevention of parenteral nutrition-associated liver disease, as measured by reduction in the incidence of cholestasis. Other clinically important benefits and harms cannot be ruled out due to low or very low certainty of evidence characterized by imprecision due to small sample sizes, inconsistencies between the studies, and risk of bias in the included studies.

Implications for research

High-certainty evidence is required to determine if enteral lipids are effective and safe in preventing and treating parenteral nutrition-associated liver disease. Well-designed and adequately powered studies should enroll infants who require parenteral nutrition for a prolonged duration and hence are at a higher risk of parenteral nutrition-associated liver disease. Such infants include those with surgical gastrointestinal diseases, short gut, and intestinal failure. Investigators should consider avoiding the use of any lipid-based preparations as placebo in the control group. A lipid-based placebo with its inherent nutritional and metabolic effect can potentially mask the specific impact of the lipid intervention and diminish the contrast between groups. Innovative ways of masking interventional lipids are required to make them indistinguishable from the control emulsion to enable blinding and decrease performance bias. Standardization of outcome definitions and measurement is paramount in improving data synthesis and interpretation.

SUPPLEMENTARY MATERIALS

Supplementary materials are available with the online version of this article: 10.1002/14651858.CD014353.pub2.

Supplementary material 1 Search strategies

Supplementary material 2 Characteristics of included studies

Supplementary material 3 Characteristics of excluded studies

Supplementary material 4 Characteristics of studies awaiting classification

Supplementary material 5 Characteristics of ongoing studies

Supplementary material 6 Analyses

Supplementary material 7 Data package

Supplementary material 8 Risk of bias

Supplementary material 9 Details of contacts with study authors

Supplementary material 10 Risk of bias domains (RoB 1)

ADDITIONAL INFORMATION

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Editorial and peer-reviewer contributions

Cochrane Neonatal supported the authors in the development of this review.

The following people conducted the editorial process for this article:

- Sign-off Editor (final editorial decision): Nai Ming Lai, School of Medicine, Taylor's University, Malaysia;
- Managing Editor (selected peer reviewers, provided editorial guidance to authors, edited the article): Hannah Payne, Cochrane Central Editorial Service;
- Editorial Assistant (conducted editorial policy checks, selected peer reviewers, collated peer-reviewer comments and supported the editorial team): Cynthia Stafford, Cochrane Central Editorial Service;
- Copy Editor (copy editing and production): Anne Lethaby, Cochrane Central Production Service;
- Peer-reviewers (provided comments and recommended an editorial decision): Christian F. Poets, MD. Dept. of Neonatology, Tuebingen University Hospital, Germany (clinical/content review); Clare Miles, Evidence Production and Methods Directorate (methods review); Jo Platt, Central Editorial information Specialist (search review).

Contributions of authors

Muralidhar Premkumar: Conceptualization, design, co-ordination of the review, search and selection of the studies for inclusion; collection of data; assessment of the risk of bias in the included studies; analysis of data; GRADE assessment; interpretation of data; writing of the review, editing of the review

Katie Huff: Co-ordination, search and selection of the studies for inclusion, collection of data for the review; assessment of the risk of bias in the included studies; analysis of data; GRADE assessment; interpretation of data; editing of the review

Chris Cooper: Search strategies, search methods, collection of data for the review

Jane Cracknell: Co-ordination of the review, writing and editing of the review

Mohan Pammi: Conception of the review; design of the review; co-ordination of the review, supervision, resolution of differences, and editing of the review

History

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Declarations of interest

Muralidhar Premkumar: A prior non-financial interest in September 2019 as part of a consultancy agreement with Fresenius Kabi, not relevant to this review. No other commercial or non-commercial conflicts of interest relevant to this review

Katie Huff: Is a prior consultant and speaker for Baxter Healthcare.

Chris Cooper: Is part of the Cochrane Neonatal group, but was not involved in the editorial review of this protocol.

Jane Cracknell: Works as a Managing Editor with the Cochrane Neonatal Group, but took no part in the editorial processing or acceptance of this review.

Mohan Pammi: Is an Editor with the Cochrane Neonatal Group, but took no part in the editorial processing or acceptance of this review.

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Registration and protocol

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Data, code and other materials

As part of the published Cochrane Review, the following are made available for download for users of the Cochrane Library: full search strategies for each database (Supplementary material 1); full citations of each unique report for all studies included, ongoing or waiting classification, or excluded at the full-text screen (Supplementary material 2; Supplementary material 3); and, in the final review; study data, including study information, study arms, and study results or test data; consensus risk of bias assessments; and analysis data, including overall estimates and settings, subgroup estimates, and individual data rows (Supplementary material 6; Supplementary material 7). Appropriate permissions have been obtained for such use. Analyses and data management were conducted within Cochrane's authoring tool, RevMan, using the inbuilt computation methods. Template data extraction forms from Covidence, Excel, etc. are available from the authors on reasonable request.

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ADDITIONAL TABLES

Table 1. Overview of included studies and synthesis table

Study ID	Type of study	Number of participants	**M: F n (%)**	**Characteristics of participants**	**Type of enteral lipid in the Intervention group**	**Type of enteral lipid in the control group**	**Duration**	**Included outcome with available data (synthesis method)**
Abou El Fadl 2021	Open-label randomized parallel-arm clinical trial	67	21:39 (35:75)	≤ 32 w GA, birthweight ≤ 1500 g	Algal oil (DHA 100 mg/day; Mom and Baby Pure DHA, Hearts and Minds PureHealth, USA)	None	14 days	Mortality: length of stay
Baack 2016	Randomized parallel-arm double-blind clinical trial	60	30:30 (50:50)	24-34 w GA	Algal oil (50 mg/day (0.18 mL) of DHA liquid; Pure Encapsulations®, Sudberg MA)	Medium-Chain Triglyceride (MCT® oil; Nestle HealthScience, Florham Park, New Jersey)	Until discharge or 37 (6/7) weeks' post-menstrual age	Feeding intolerance; central line-associated bloodstream infections: mortality
Bernabe-Garcia 2019	Randomized parallel-arm double-blind clinical trial	136	53: 57 (48:52)	< 1500 g but ≥ 1000 g with functional gastrointestinal tract	Algal oil (75 mg of DHA/kg) diluted in high oleic sunflower oil as vehicle (life's DHA, DSM Nutritional Products Inc., Parsippany, NJ, USA)	Sham oil (high oleic sunflower oil prepared by PROGELA SA, Mexico City, Mexico).	14 days	Feeding intolerance; mortality
Bernabe-García 2021	Randomized double-blind parallel-arm study	214	99:101 (49:51)	1000-1500 g birthweight with functional gastrointestinal tract	Algal oil (75 mg of DHA/kg) diluted in high oleic sunflower oil as vehicle (Neuromins® for Kids life's DHA®; DSM Nutritional Products Inc., Parsippany, NJ, USA)	Sham oil (high oleic sunflower oil prepared by PROGELA SA, Mexico City, Mexico).	14 days	Mortality
Collins 2017	Randomized double-blind parallel-arm study	1273	680:593 (53:47)	< 29 w GA	Fractionated tuna oil 0.5 mL/kg (70% of total oil as DHA in triglyceride form; 0.5 mL contained 60 mg of DHA, 4 mg of eicosapentaenoic acid, 4.6 mg of vitamin C, and 0.05 mg of vitamin E)	Soy emulsion 0.5 mL/kg (0.5 ml contained 55 mg of linoleic acid, 5.7 mg of alpha-linolenic	Until 36 w post-menstrual age	Central line-associated bloodstream infections; mortality; time to full feeds; growth (z-scores)

Table 1. Overview of included studies and synthesis table *(Continued)*

						acid, 4.8 mg of vitamin C, and 0.01 mg of vitamin E)		
Yang 2014	Open-label randomized parallel-arm clinical trial	37	23:13 (64:36)	Premature infants < 2 months of age with jejunostomy or ileostomy. Both before and after re-anastomosis	Fish oil 2 g/day (Major fish oil 500, Major Pharmaceuticals, Livonia, Michigan; Rugby Sea Omega 50, Rugby Pharmaceuticals Inc., Corona, California) and Microlipid (safflower oil; Nestle Nutrition, Florham Park, New Jersey)	None	Until discharge in infants with short bowel syndrome, or until full feeds were achieved in infants with conjugated bilirubin < 2 mg/dL	Central line-associated bloodstream infections; mortality
NCT01674478	Open-label randomized parallel-arm clinical trial	18	11:7 (61:59)	Infants who were ≤ 2-months-old with birthweight ≤ 1250 g with an enterostomy	Microlipid with fish oil	Microlipid	Within 4 days from placement of the enterostomy until reanastomosis.	Mortality
Frost 2021	Randomized double-blind parallel-arm study	30	12:18 (40:60)	Very low birthweight infants (< 1500 g birthweight)	Algal and fungal oil (LCPUFA-120 (DHA 40, AA 80), LCPUFA-360 (DHA 120, AA 240)	Sunflower oil	For 8 weeks or discharge whichever was earlier	Prevention of liver disease (incidence of cholestasis); feeding intolerance; mortality
Hellstrom 2021	Open-label randomized parallel-arm clinical trial	206	117:89 (57:43)	< 28 weeks GA	Fungal oil (AA 100) + Algal oil (DHA 50) (Formulaid, DSM Nutritional Products Inc)	None	Until 40 weeks' postmenstrual age	Prevention of liver disease (incidence of cholestasis); mortality
Wendel 2023 [57, 58]	Randomized double-blind parallel-arm study	121	66:54 (55:45)	< 29 w GA	Algal oil (DHA 50 mg/kg) and fungal oil (AA 100 mg/kg) (Formulaid™, DSM Nutritional Products Inc.)	Medium-chain triglycerides (MCT-oil™, Nutricia)	Until 36 w postmenstrual age	Central line-associated bloodstream infections & mortality (Wendel K 2023);

Table 1. Overview of included studies and synthesis table *(Continued)*

								growth (z-scores) (Rosshold 2023)
Robinson 2016	Randomized double-blind parallel-arm study	30	17:11 (61:39)	< 34 weeks, < 1000 g birthweight	Algal oil + fungal oil in oleic sunflower oil (low-dose DHA 20 mg/kg + AA 40 mg/kg; high-dose DHA 60 mg/kg + AA 120 mg/kg; DSM Nutritional Products)	Oleic sunflower oil	Until 8 weeks of age or discharge whichever was earlier	Prevention of liver disease (incidence of cholestasis); central line-associated bloodstream infections; mortality

AA; arachidonic acid; **DHA**: docosahexaenoic acid; **F**: female; **GA**: gestational age; **LCPUFA**:long chain polyunsaturated fatty acids; **M**: male; **MCT**: medium-chain triglyceride; **W**: weeks